Computers and Fluids 219 (2021) 104873

Contents lists available at ScienceDirect

Computers and Fluids

journal homepage: www.elsevier.com/locate/compfluid

Simplified artificial viscosity approach for curing the shock instability

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ARTICLE INFO

Article history:
Received 3 September 2020
Revised 14 December 2020
Accepted 3 February 2021
Available online 10 February 2021

Keywords:
Carbuncle phenomenon
Shock instability
Godunov-type methods
Riemann solvers
Artificial viscosity

ABSTRACT

The artificial viscosity approach for curing the carbuncle phenomenon and suppressing post-shock oscillations has recently been presented and successfully tested on both first-order and high-order Godunov-type schemes. This approach introduces some dissipation in the form of the right-hand side of Navier–Stokes equations into the basic method of solving Euler equations. The current study presents a simplified form of the artificial viscosity that is comparable to the original one in its efficiency. We also discuss the distinctive features of the artificial viscosity approach as compared to other ways of curing the carbuncle phenomenon. In addition, the accuracy of several approximate Riemann solvers has been examined in combination with the artificial viscosity approach.

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1. Introduction

The carbuncle phenomenon represents a numerical instability that affects the capturing of strong shocks when using Godunov-type schemes [1,2]. This instability can reveal itself either by saw-tooth perturbations or by unphysical kinks of the shock front. The instability arises if the following conditions are meet: (1) the physical viscosity is zero or its influence on smearing the shock is negligibly small (compare to the numerical viscosity); (2) the shock Mach number, M_S, exceeds some threshold value that depends on the numerical scheme and the problem specification; (3) the grid (one family of grid lines) aligns with the flow lines at the shock front; (4) the numerical scheme employs a Riemann solver that can capture a contact discontinuity (a complete Riemann solver). The common practice for curing the carbuncle instability is to increase the scheme dissipation (the numerical viscosity) within the shock layer by modifying the Riemann solver in use (see summaries on this subject in [3–8]).

In [7] the author proposed a novel approach for curing the problem. Its idea is to introduce some dissipation in the form of the right-hand side of Navier-Stokes equations into the basic method of solving Euler equations; in so doing, the molecular viscosity coefficient is replaced by the artificial viscosity coefficient. The new cure for the carbuncle flaw, the artificial viscosity (AV) approach, has first been tested and tuned for the case of using first-order schemes in two-dimensional simulations. Later, in [9] the approach was extended to the case of three-dimensional simulations, and in [10] it was adopted for the high-order Godunov-type schemes. The efficiency of the AV approach has been demonstrated on a variety of well-known test problems.

Providing a broad-spectrum beneficial effect, the AV approach is versatile: it is external to the inviscid flux solver, and it can be used in CFD codes without switching or tuning to any specific problem. In a shock-free problem it will have no impact on the solution; in a problem with shock(s) it will assure the necessary dissipation in the shock layer to prevent the carbuncle instability and improve the post-shock solution. Considering that many of the existing CFD codes support simulations in the framework of both Euler and Navier-Stokes equations, the AV approach could readily be implemented within such codes. However, introducing this approach into the codes developed solely for solving the Euler equations may seem to be over-complicated.

In this work we present a *simplified* form of the AV approach that is comparable to the original one in its efficiency. The paper is organized as follows. The next section (Section 2) describes the schemes selected for the study and the AV approach in both its original and simplified forms. Section 3 demonstrates the efficiency of the simplified AV approach on various test problems. In Section 4 we discuss the distinctive features of the AV approach in comparison with other ways of curing the carbuncle phenomenon. In Section 5 we examine the accuracy of several approximate Riemann solvers in combination with the artificial viscosity approach. Finally, conclusions are given in Section 6.

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https://doi.org/10.1016/j.compfluid.2021.104873
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2. Numerical schemes

In this study we restrict our attention to the class of high-order schemes as those having the highest practical significance. More specifically, we will deal with the schemes that were selected in [10] for testing the original AV approach. So in this section we first briefly recall the description of the selected schemes and then specify the AV approaches (the original and simplified ones) in more detail.

2.1. The Euler equations

We describe the schemes in the context of solving two-dimensional equations of gas dynamics, which in Cartesian coordinates xy read

$$\frac{\partial \mathbf{U}}{\partial t}+\frac{\partial \mathbf{F}_x}{\partial x}+\frac{\partial \mathbf{F}_y}{\partial y}=\mathrm{RHS},$$
$$\mathbf{U}=\begin{bmatrix}\rho\\ \rho u_x\\ \rho u_y\\ \rho h_0-p\end{bmatrix},\quad \mathbf{F}_x=\begin{bmatrix}\rho u_x\\ \rho u_x^2+p\\ \rho u_x u_y\\ \rho u_x h_0\end{bmatrix},\quad \mathbf{F}_y=\begin{bmatrix}\rho u_y\\ \rho u_x u_y\\ \rho u_y^2+p\\ \rho u_y h_0\end{bmatrix}, \tag{1}$$

where RHS $=0$ for the Euler equations, $\mathbf{u}=(u_x,u_y)$ are the velocity vector and its components, ρ is the density, p is the pressure, and h_0 is the specific total enthalpy, which for a polytropic gas is given by

$$h_0=\frac{1}{2}(u_x^2+u_y^2)+h,\quad h=\frac{\gamma p}{(\gamma-1)\rho},$$

with h denoting the specific enthalpy and γ denoting the ratio of specific heats.

2.2. Space-time discretization

We apply the finite volume approach to solve (1) numerically. In so doing we consider a sufficiently smooth structured grid and introduce curvilinear coordinates $\xi\eta$ that transform the grid in physical space xy to a rectangular grid in computational space with the grid spacing $\Delta\xi=\Delta\eta=1$. The following geometric parameters of the grid will take the place of metrics (hereinafter the grid indices i and j correspond to the coordinates ξ and η): $V_{i,j}$, the cell volume; $(\mathbf{S}_\xi)_{i+½,j}=(S_{\xi x},S_{\xi y})_{i+½,j}$ and $(\mathbf{S}_\eta)_{i,j+½}=(S_{\eta x},S_{\eta y})_{i,j+½}$, the area vectors of the cell faces in two grid directions (each area vector points in the direction of increase in the associated coordinate); see Fig. 1.

Assume that the solution at a given time level n is defined, meaning that for the time $t=t^n$ we know the average values $\mathbf{U}^n_{i,j}$ within each cell. In order to update the solution to the next time level $n+1$ we integrate (1) over the cell and the time between t^n and $t^{n+1}\equiv t^n+\Delta t$. The result is

$$\mathbf{U}^{n+1}_{i,j}=\mathbf{U}^n_{i,j}-\frac{\Delta t}{V_{i,j}}\Big[\big(\mathbf{F}_\xi\big)_{i+1/2,j}-\big(\mathbf{F}_\xi\big)_{i-1/2,j}+\big(\mathbf{F}_\eta\big)_{i,j+1/2}-\big(\mathbf{F}_\eta\big)_{i,j-1/2}\Big],$$

where $\mathbf{F}_\xi=S_{\xi x}\mathbf{F}_x+S_{\xi y}\mathbf{F}_y$ and $\mathbf{F}_\eta=S_{\eta x}\mathbf{F}_x+S_{\eta y}\mathbf{F}_y$ are the flux vectors across the cell faces in two grid directions. The way of computing fluxes depends on the specific scheme. Thus for the first-order Godunov-type schemes the fluxes are computed by one of the Riemann solvers and can be expressed as

$$\big(\mathbf{F}_\xi\big)_{i+1/2,j}=\mathbf{F}^{RS}\Big(\mathbf{Q}^n_{i,j},\mathbf{Q}^n_{i+1,j},\big(\mathbf{S}_\xi\big)_{i+1/2,j}\Big),$$
$$\big(\mathbf{F}_\eta\big)_{i,j+1/2}=\mathbf{F}^{RS}\Big(\mathbf{Q}^n_{i,j},\mathbf{Q}^n_{i,j+1},\big(\mathbf{S}_\eta\big)_{i,j+1/2}\Big),$$

where $\mathbf{Q}\equiv[u_x,u_y,\rho,p]^T$ and $\mathbf{Q}^n_{i,j}=\mathbf{Q}(\mathbf{U}^n_{i,j})$.

2.3. Reconstructions and time stepping

In the second-order MUSCL-type schemes the piecewise linear distributions of flow variables are reconstructed based on the known values of $\mathbf{Q}^n_{i,j}$ at the cell centers. On a smooth structured grid the reconstruction process reduces to calculating the slopes, $\Delta_\xi\mathbf{Q}^n_{i,j}$ and$\Delta_\eta\mathbf{Q}^n_{i,j}$. The boundary extrapolated values are then found as

$$\mathbf{Q}^n_{i+1/2-,j}=\mathbf{Q}^n_{i,j}+\frac{1}{2}\Delta_\xi\mathbf{Q}^n_{i,j},\quad \mathbf{Q}^n_{i-1/2+,j}=\mathbf{Q}^n_{i,j}-\frac{1}{2}\Delta_\xi\mathbf{Q}^n_{i,j},$$
$$\mathbf{Q}^n_{i,j+1/2-}=\mathbf{Q}^n_{i,j}+\frac{1}{2}\Delta_\eta\mathbf{Q}^n_{i,j},\quad \mathbf{Q}^n_{i,j-1/2+}=\mathbf{Q}^n_{i,j}-\frac{1}{2}\Delta_\eta\mathbf{Q}^n_{i,j},$$

and for the fluxes we can write

$$\big(\mathbf{F}_\xi\big)_{i+1/2,j}=\mathbf{F}^{RS}\Big(\mathbf{Q}^n_{i+1/2-,j},\mathbf{Q}^n_{i+1/2+,j},\big(\mathbf{S}_\xi\big)_{i+1/2,j}\Big),$$
$$\big(\mathbf{F}_\eta\big)_{i,j+1/2}=\mathbf{F}^{RS}\Big(\mathbf{Q}^n_{i,j+1/2-},\mathbf{Q}^n_{i,j+1/2+},\big(\mathbf{S}_\eta\big)_{i,j+1/2}\Big).$$

The slopes are computed by a slope-limiter algorithm for each grid direction independently. Hereinafter we consider three slope limiters (named in accordance with the book [11]):

$$\Delta f_i=\mathrm{minmod}\big(\Delta f_{i-1/2},\Delta f_{i+1/2}\big),$$
$$\mathrm{minmod}(a,b)\equiv\begin{cases}a, & \text{if } |a|\le|b| \text{ and } ab>0,\\ b, & \text{if } |b|<|a| \text{ and } ab>0,\\ 0, & \text{if } ab\le 0,\end{cases}$$

$$\Delta f_i=\mathrm{vanLeer}\big(\Delta f_{i-1/2},\Delta f_{i+1/2}\big),$$
$$\mathrm{vanLeer}(a,b)\equiv\begin{cases}\frac{2ab}{a+b}, & \text{if } ab>0,\\ 0, & \text{if } ab\le 0,\end{cases}$$

$$\Delta f_i=\mathrm{MC}\big(\Delta f_{i-1/2},\Delta f_{i+1/2}\big),$$
$$\mathrm{MC}(a,b)\equiv\mathrm{minmod}\left(\frac{a+b}{2},2\mathrm{minmod}(a,b)\right).$$

where $\Delta f_{i-1/2}=f_i-f_{i-1}$ and $\Delta f_{i+1/2}=f_{i+1}-f_i$.

In our study we apply reconstructions based on *characteristic variables* that allow reducing the post-shock oscillations as opposed to *primitive variables* (components of the vector $\mathbf{Q}$). See the previous study [10] for details.

In addition to the piecewise linear reconstructions, we employ the fifth-order reconstruction WENO5, as it was described in [12].

To increase the accuracy in time we use: (a) the second- and third-order Runge-Kutta methods [13] (RK2 and RK3, respectively), and (b) a special predictor-corrector technique that we call the Hancock-Rodionov (HR) scheme (see [10] for details). Notice that being second-order accurate the HR scheme requires less computing time and usually provides better accuracy than the RK2 method.

2.4. The Navier-Stokes equations

The right-hand side of the Navier-Stokes equations reads

$$\mathrm{RHS}=\frac{\partial \mathbf{F}^v_x}{\partial x}+\frac{\partial \mathbf{F}^v_y}{\partial y},\quad \mathbf{F}^v_m=[0,\tau_{xm},\tau_{ym},q_m+u_x\tau_{xm}+u_y\tau_{ym}]^T,$$
$$m=x,y, \tag{2}$$

where the stress tensor (τ_{nm}) and the heat flux (q_m) can be written as

$$\tau_{nm}=\mu\left[\frac{\partial u_n}{\partial m}+\frac{\partial u_m}{\partial n}-\frac{2}{3}(\nabla\mathbf{u})\delta_{nm}\right],\quad \nabla\mathbf{u}=\frac{\partial u_x}{\partial x}+\frac{\partial u_y}{\partial y},$$
$$q_m=\lambda\frac{\partial T}{\partial m}=\frac{\mu}{\mathrm{Pr}}\left(\frac{\partial h}{\partial m}\right),\quad m,n=x,y.$$

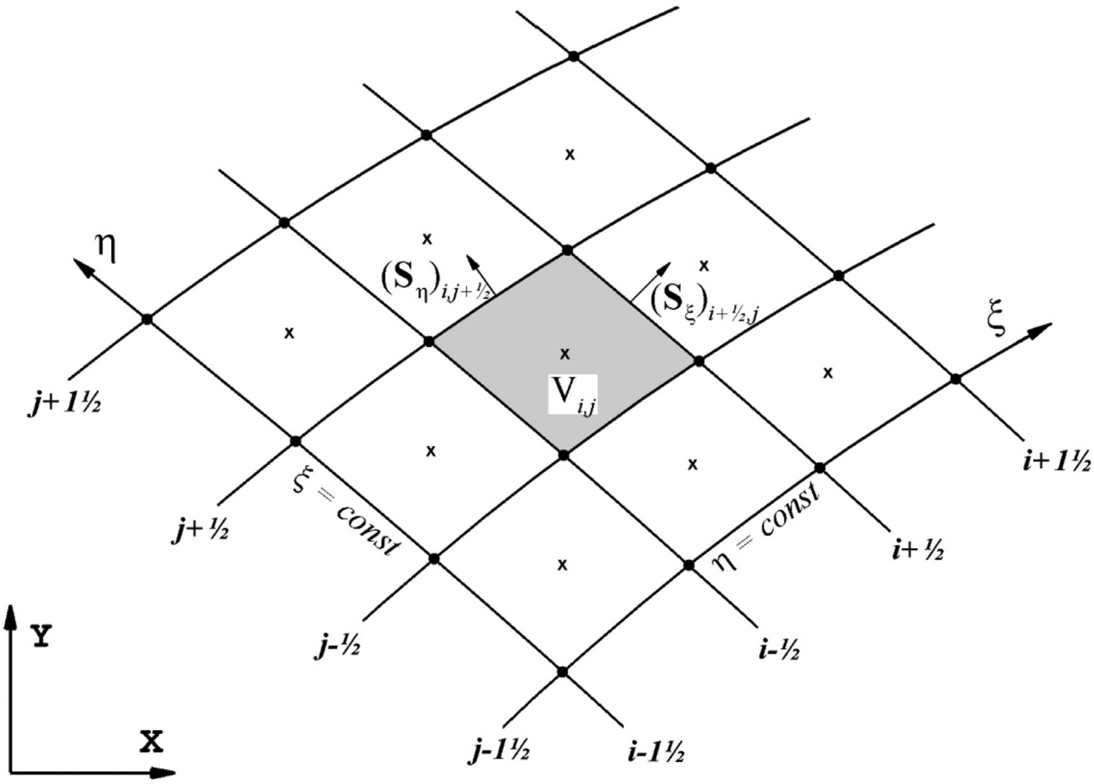

Fig. 1. Discretization of two-dimensional physical space with smooth structured grid.

Here μ is the molecular viscosity coefficient, Pr is the Prandtl number and δ_{mn} is the Kronecker delta.

The space discretization of (2) takes the form

$$\mathrm{RHS} = \frac{1}{V_{i,j}}\left[\left(\mathbf{F}_\xi^{\mathrm{v}}\right)^n_{i+1/2,j} - \left(\mathbf{F}_\xi^{\mathrm{v}}\right)^n_{i-1/2,j} + \left(\mathbf{F}_\eta^{\mathrm{v}}\right)^n_{i,j+1/2} - \left(\mathbf{F}_\eta^{\mathrm{v}}\right)^n_{i,j-1/2}\right],$$

where $\mathbf{F}_\xi^{\mathrm{v}} = S_{\xi x}\mathbf{F}_x^{\mathrm{v}} + S_{\xi y}\mathbf{F}_y^{\mathrm{v}}$ and $\mathbf{F}_\eta^{\mathrm{v}} = S_{\eta x}\mathbf{F}_x^{\mathrm{v}} + S_{\eta y}\mathbf{F}_y^{\mathrm{v}}$. To compute viscous fluxes we use explicit central difference approximations and the following transformation formulae

$$\frac{\partial f}{\partial x} = \frac{1}{V}\left(S_{\xi x}\frac{\partial f}{\partial \xi} + S_{\eta x}\frac{\partial f}{\partial \eta}\right), \quad \frac{\partial f}{\partial y} = \frac{1}{V}\left(S_{\xi y}\frac{\partial f}{\partial \xi} + S_{\eta y}\frac{\partial f}{\partial \eta}\right),$$
$$f = u_x, u_y, h.$$

Thus the viscous flux vector in a particular grid direction (ξ-direction, for instance) can be written as

$$\mathbf{F}_\xi^{\mathrm{v}} = S_{\xi x}\mathbf{F}_x^{\mathrm{v}} + S_{\xi y}\mathbf{F}_y^{\mathrm{v}} = \left[0, \tau_{x\xi}, \tau_{y\xi}, q_\xi + u_x\tau_{x\xi} + u_y\tau_{y\xi}\right]^T, \tag{3}$$

with

$$\tau_{x\xi} = \frac{\mu}{V}\left[\left(\frac{4}{3}S_{\xi x}^2 + S_{\xi y}^2\right)\frac{\partial u_x}{\partial \xi} + \left(\frac{4}{3}S_{\xi x}S_{\eta x} + S_{\xi y}S_{\eta y}\right)\frac{\partial u_x}{\partial \eta}\right.$$
$$\left. + \frac{1}{3}S_{\xi x}S_{\xi y}\frac{\partial u_y}{\partial \xi} + \left(S_{\xi y}S_{\eta x} - \frac{2}{3}S_{\xi x}S_{\eta y}\right)\frac{\partial u_y}{\partial \eta}\right],$$

$$\tau_{y\xi} = \frac{\mu}{V}\left[\frac{1}{3}S_{\xi x}S_{\xi y}\frac{\partial u_x}{\partial \xi} + \left(S_{\xi x}S_{\eta y} - \frac{2}{3}S_{\xi y}S_{\eta x}\right)\frac{\partial u_x}{\partial \eta}\right.$$
$$\left. + \left(S_{\xi x}^2 + \frac{4}{3}S_{\xi y}^2\right)\frac{\partial u_y}{\partial \xi} + \left(S_{\xi x}S_{\eta x} + \frac{4}{3}S_{\xi y}S_{\eta y}\right)\frac{\partial u_y}{\partial \eta}\right],$$

$$q_\xi = \frac{\mu}{\mathrm{Pr}\,V}\left[\left(S_{\xi x}^2 + S_{\xi y}^2\right)\frac{\partial h}{\partial \xi} + \left(S_{\xi x}S_{\eta x} + S_{\xi y}S_{\eta y}\right)\frac{\partial h}{\partial \eta}\right].$$

To approximate the flux vector components at the cell interface (i+½, j) we use

$$S_{\xi x} = \left(S_{\xi x}\right)_{i+1/2,j},$$
$$S_{\eta x} = \frac{1}{4}\left[\left(S_{\eta x}\right)_{i,j-1/2} + \left(S_{\eta x}\right)_{i+1,j-1/2} + \left(S_{\eta x}\right)_{i,j+1/2} + \left(S_{\eta x}\right)_{i+1,j+1/2}\right],$$

$$S_{\xi y} = \left(S_{\xi y}\right)_{i+1/2,j},$$
$$S_{\eta y} = \frac{1}{4}\left[\left(S_{\eta y}\right)_{i,j-1/2} + \left(S_{\eta y}\right)_{i+1,j-1/2} + \left(S_{\eta y}\right)_{i,j+1/2} + \left(S_{\eta y}\right)_{i+1,j+1/2}\right], \tag{4}$$

$$\frac{\partial f}{\partial \xi} = f_{i+1,j} - f_{i,j}, \quad \frac{\partial f}{\partial \eta} = \frac{1}{4}\left(f_{i,j+1} + f_{i+1,j+1} - f_{i,j-1} - f_{i+1,j-1}\right),$$
$$f = u_x, u_y, h.$$

Once the components of RHS are computed at a given time level n for each cell, these values remain frozen (not updated) at all stages of marching the solution to the next time level n+1.

2.5. The original AV approach

In the original AV approach [7], although simulating inviscid problems, we introduced additional dissipation in the form of right-hand sides of the Navier-Stokes equations (see Section 2.4) where the molecular viscosity coefficient (μ) is replaced by the artificial viscosity coefficient (μ_{AV}); the Prandtl number has been set to ¾. For the artificial viscosity coefficient we chose the formula

$$\mu_{AV} = \begin{cases} C_{AV}\rho h^2\sqrt{(\nabla\mathbf{u})^2 - (C_{th}a/h)^2}, & \text{if } -(\nabla\mathbf{u}) > C_{th}a/h, \\ 0, & \text{otherwise}, \end{cases} \tag{5}$$

where a is the sound speed, h is the characteristic mesh size, C_{AV} is the dimensionless parameter, $C_{th} = 0.05$ is the coefficient in the compression intensity threshold that restricts the effect of artificial viscosity to the shock layer only.

In the two-dimensional case with non-square cells the characteristic mesh size is computed as $h = \max(d_1, d_2)/\sqrt{2}$, with d_1 and d_2 denoting the lengths of the cell diagonals. For rectangular cells in Cartesian coordinates it reduces to $h = \sqrt{(h_x^2 + h_y^2)/2}$. In the three-dimensional case we use the expression for h that was derived in [9].

Initially the artificial viscosity coefficient is defined at the cell centers, and its values at the cell interfaces are then found by linear interpolation: $f_{i+½,j} = 0.5(f_{i,j} + f_{i+1,j})$. The velocity vector divergence in (5) is calculated by Gauss's theorem; in so doing the values of the velocity vector components at the cell interfaces are also found by linear interpolation between the cell center values.

With introducing the viscous terms (2) into the Euler equations (which are the convection equations of hyperbolic type) we actually turn to solving convection–diffusion equations of parabolic type. In this case we should account for an additional constraint on the time step Δt, which can be done by

$$\Delta t = C_{cfl} \cdot \min_{i,j}\left\{\left[\frac{1}{(\Delta t^{conv})_{i,j}} + \frac{1}{\left(\Delta t^{diff}\right)_{i,j}}\right]^{-1}\right\},$$
$$\Delta t^{conv} = \frac{V}{(\mathbf{u}{\cdot}\mathbf{S}_\xi) + (\mathbf{u}{\cdot}\mathbf{S}_\eta) + a\left(S_\xi + S_\eta\right)}, \quad \Delta t^{diff} = \frac{3\rho}{8\mu_{AV}} \cdot \frac{V^2}{S_\xi^2 + S_\eta^2},$$

where C_{cfl} is the Courant number ($C_{cfl} < 1$), $S_\xi = |\mathbf{S}_\xi|$, $S_\eta = |\mathbf{S}_\eta|$. Notice that the artificial viscosity is introduced in the shock layer only and its effect on a single fluid particle is short-time. Our experience reveals that in some cases the stable computation is still possible without additional constraint (when $\Delta t^{diff} = \infty$). Nonetheless, in this paper we only report the data obtained with it.

For the principal coefficient of the AV approach we have chosen the value of C_{AV}= 0.5 that ensures the suppression of shock instability if two recommendations for the high-order schemes are followed [10]. The first recommendation relates to the procedure of data reconstruction; it is essential to use the characteristic variables here. As stated above, we apply this recommendation throughout our study. The second recommendation refers to the data reconstruction in the cells that fall inside the shock layer. One should rest upon the concept of piecewise linear distribution of the data within such cells and use the minmod limiter (see [10] for details). To denote the latter technique, we add the character (m) to the basic reconstruction, for example: MC(m).

Remark. Although in most test cases a smaller value of C_{AV}, say C_{AV}= 0.3, was sufficient to suppress the carbuncle instability, we recommend using the value C_{AV}= 0.5 for two reasons: (1) it is all-purpose; (2) reducing C_{AV} will not make the shock layer significantly thinner, but it may increase the level of post-shock oscillations.

2.6. Simplified AV approach

The artificial viscosity acts within a thinly smeared shock, the thickness of which reduces directly with grid refinement. It differs fundamentally from smearing by the physical viscosity, when we *approximate* the Navier-Stokes equations within the shock layer. The dominant cause of following the exact form of Eqs. (2) in the AV approach is that it can be easily implemented in a rich variety of CFD codes that provide solve both the Euler and the Navier-Stokes equations. However, introducing this form into the codes developed solely for solving the Euler equations may seem to be over-complicated. Actually, it was found that the AV approach can be considerably simplified without reducing its efficiency. The main principles that we followed in the process of modifying the AV approach were as follows.

1. Like the original equations for the viscous flux vector in the ξ–th grid direction (3), the simplified equations should not depend on the orientation of the shock front. In other words, the algorithm for calculating the viscous flux vector must be the same for all the cell faces, such as those located in the plane of the shock front, or perpendicular to it.

2. When simplifying the AV approach, it is extremely important to reduce the stencil for calculating the viscous flux vector. Therefore, we will discard all terms containing derivatives in the η-th grid direction from the components of the vector (3).

3. The next stage of modification is to follow the gradient approach, when $\tau_{x\xi}$ depends only on the gradient of u_x, and $\tau_{y\xi}$ – on the gradient of u_y.

4. Finally, in the last component of the vector $\mathbf{F}_\xi^{\mathrm{v}}$ (energy flux), we leave only the dependence on the enthalpy gradient. Although this component is not required for curing the carbuncle instability, it is beneficial (at least in a simplified form) for another reason: the use of artificial viscosity without heat conductivity intensifies the entropy trace.

Thus, in the simplified artificial viscosity approach (sAV), we do not follow the form of Eqs. (2); instead, we exclude all but the most necessary components directly from the flux vector (3). The resulting expression for the intercell flux is

$$\mathbf{F}_\xi^{\mathrm{v}} = S_{\xi x}\mathbf{F}_x^{\mathrm{v}} + S_{\xi y}\mathbf{F}_y^{\mathrm{v}} = \left[0,\, \hat{\tau}_{x\xi},\, \hat{\tau}_{y\xi},\, \hat{q}_\xi\right]^T, \tag{6}$$

with

$$\hat{\tau}_{x\xi} = \frac{\mu S_\xi^2}{V}\frac{\partial u_x}{\partial \xi}, \quad \hat{\tau}_{y\xi} = \frac{\mu S_\xi^2}{V}\frac{\partial u_y}{\partial \xi}, \quad \hat{q}_\xi = \frac{\mu S_\xi^2}{V}\frac{1}{\mathrm{Pr}}\frac{\partial h}{\partial \xi}.$$

Then, at the cell interface (i+½, j) we only need the following simple approximations

$$\frac{\mu S_\xi^2}{V} = \frac{\left(\mu_{i+1,j} + \mu_{i,j}\right)}{\left(V_{i+1,j} + V_{i,j}\right)}\left|\left(\mathbf{S}_\xi\right)_{i+1/2,j}\right|^2,$$
$$\frac{\partial f}{\partial \xi} = f_{i+1,j} - f_{i,j}, \quad f = u_x, u_y, h. \tag{7}$$

All other elements of the AV approach (stated in Section 2.5) are retained in the sAV approach without modification; namely: formulas for calculating the coefficient μ_{AV} in the cell centers (including the values of its constants) and for its interpolation on the cell interfaces; the additional constraint on the time step Δt; all recommendations for using the artificial viscosity approach in high-order schemes.

3. Testing the sAV appsroach

The simplified AV approach was tested using the schemes described in the previous section: HR-minmod, HR-vanLeer, HR-MC, RK2-MC and RK3-WENO5 (again, these schemes were used in [10] for testing the original AV approach). The Riemann problem was generally solved by the exact (Godunov) solver [14]; sample computations with the Roe [15] and HLLC [16] solvers gave close results (like in the case of using the original AV approach). Computations by the RK3-WENO5 scheme were done with the Courant number C_{cfl} = 0.6; for all the other schemes we took C_{cfl} = 0.8.

Before demonstrating the results of testing the sAV approach on carbuncle-related problems, we will make two comments about the use of this approach in carbuncle-free problems. First, the algorithm for calculating the coefficient μ_{AV} in the sVA approach is the same as in the original AV approach. Therefore, in shock-free problems the sVA approach, as well as the AV approach, does not affect the solution. In [10] we demonstrated the properties of the selected schemes (including the convergence study) on one of the problems of this class (see Appendix A in [10]). Second, the use of the AV approach (either in the original or simplified version) can be very useful in problems with shock waves of moderate intensity, when the carbuncle instability not manifested. For example, in Appendix A we demonstrate this kind of case by solving a test problem of the inviscid strong vortex-shock wave interaction at M_S = 1.5.

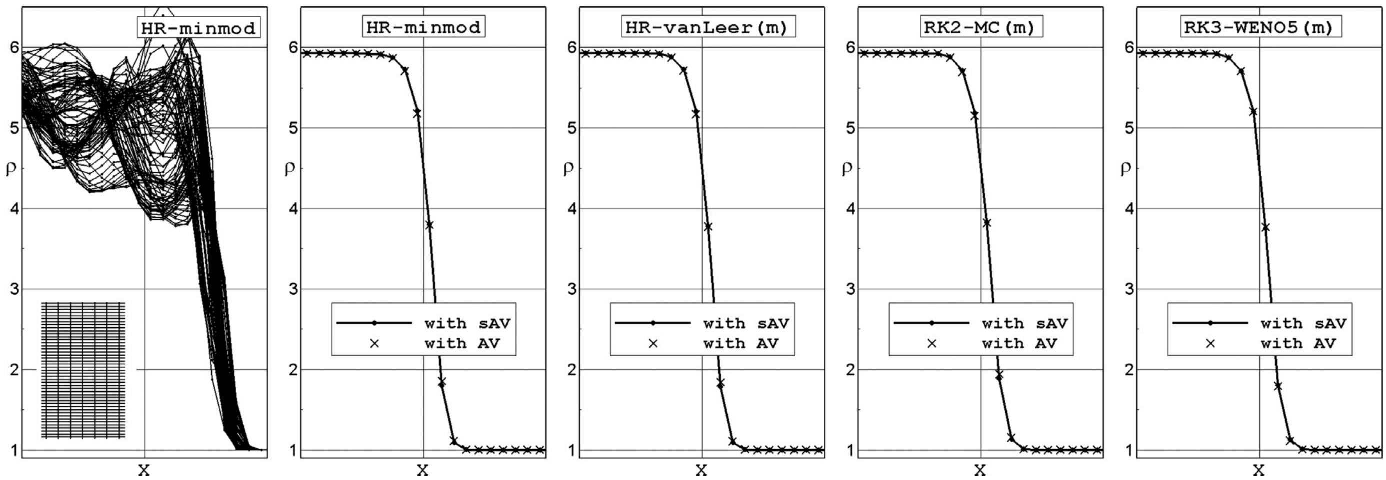

Fig. 2. Advancing-shock-wave test computations by different schemes on the rectangular grid with $h_y/h_x = ¼$. The density profiles obtained without artificial viscosity (left plot) and with the sAV and AV approaches (other plots).

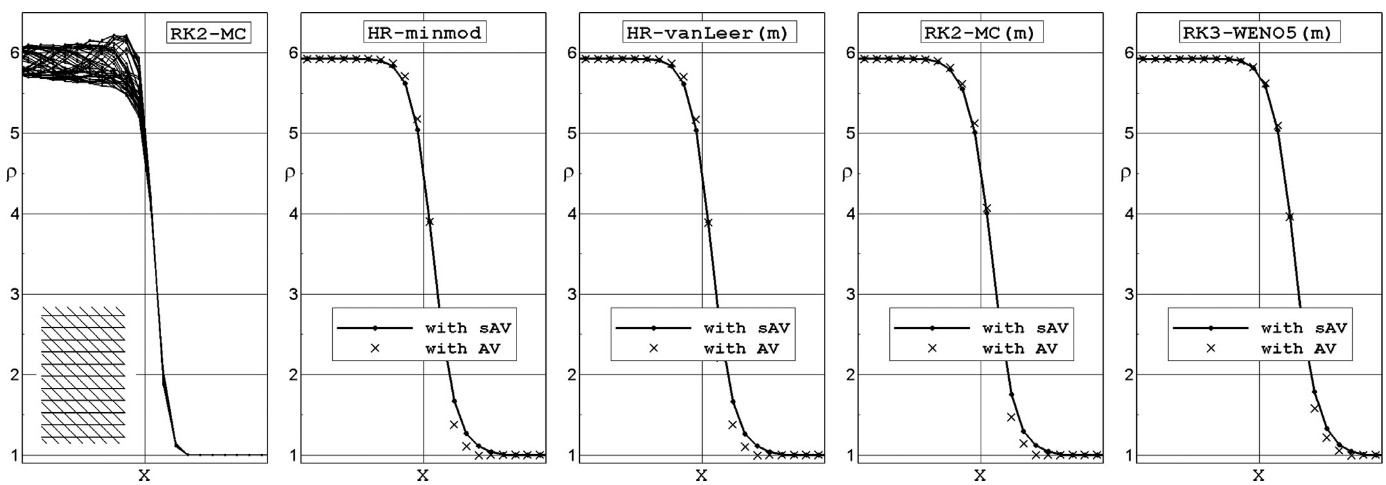

Fig. 3. Advancing-shock-wave test computations by different schemes on the parallelogram grid with $h_y/h_x = 1$ and $\alpha = 1$. The density profiles obtained without artificial viscosity (left plot) and with the sAV and AV approaches (other plots).

3.1. Quirk-type test problems

First of all we have considered the Quirk-type test problems as they were formulated in [10]. All of these tests serve to check the stability of a planar shock wave, advancing or steady, during its computation in the two-dimensional case. It has been found previously that the most prominent shock instability is observed in hypersonic flows. Thus for all test problems we have assumed that the shock-wave Mach number $M_S = 20$.

The results of our study demonstrated that the ability of the simplified AV approach to suppress the shock instability in the Quirk-type tests is comparable with that of the original AV approach. Figs. 2–5 illustrate this statement for the following test cases: (1) the advancing-shock-wave test problem on the rectangular grid with $h_y/h_x = ¼$; (2) the advancing-shock-wave test problem on the parallelogram grid with $h_y/h_x = 1$ and $\alpha = 1$ (the vertical grid lines $x =$ const are replaced with slanting lines $x + \alpha y =$ const); (3) the quasisteady-shock-wave test problem on the rectangular grid with $h_y/h_x = ½$; (4) the steady-shock-wave test problem on the parallelogram grid with $h_y/h_x = ½$ and $\alpha = 0.02$. See [10] for more details on the Quirk-type test formulations. The left plots in each of the figures show the solutions without adding the artificial viscosity; these plots also include fragments of the grids.

3.2. Other carbuncle-related tests

Following the procedure of our previous study we have checked the efficiency of the sAV approach on other carbuncle-related tests: (1) the double Mach reflection problem, (2, 3) super- and hypersonic flow around a cylinder, (4) hypersonic flow around a sphere (3D case), (5) the Sedov blast wave problem and (6) the Noh problem (see [10] for more details on the formulations of the tests). In so doing we have applied the selected schemes in conjunction with various Riemann solvers (RS).

Fig. 6 shows the results of computations of the double Mach reflection problem obtained by the RK3-WENO5 scheme, both in the original version and with adding the AV and sAV approaches. As one can see, when the original RK3-WENO5 scheme is used (upper plot), the solution is highly distorted by the carbuncle phenomenon and post-shock oscillations. Adding the artificial viscosity approach (middle and bottom plots) remedies the defects of the solution completely; as this takes place, both versions of the approach (original and simplified) show almost identical results.

Similar comparison results were obtained for other tests. For the sake of reducing the volume of this article, Figs. 7–11 display only results of our present study (data obtained with the addition of the sAV approach). We encourage the reader to compare them with the corresponding data in [10]: Figs. 11, 12, 15–17 (they were

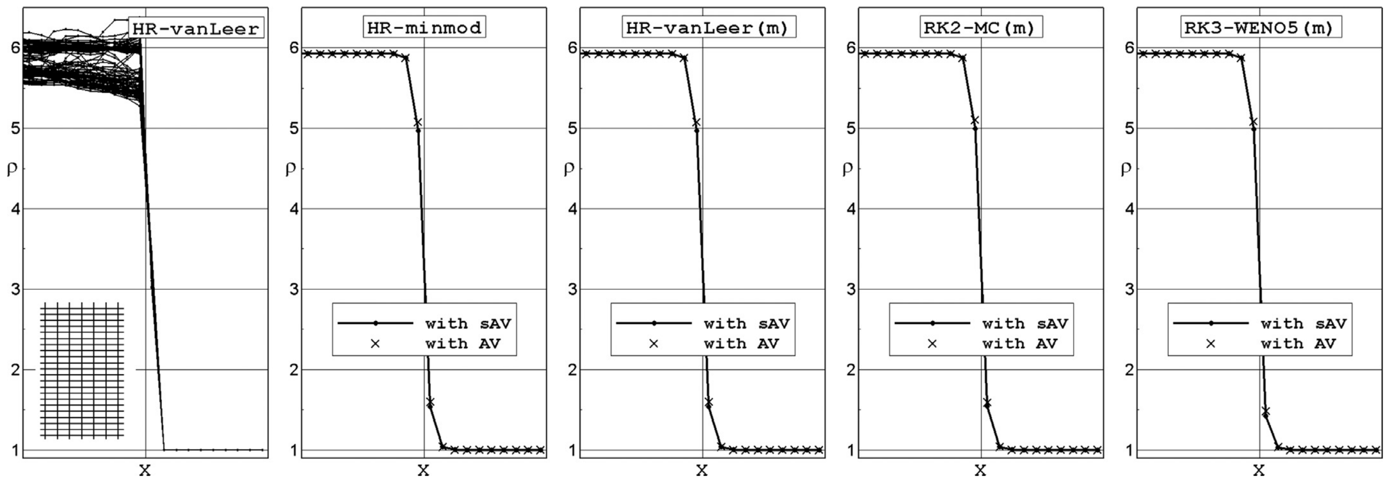

Fig. 4. Quasisteady-shock-wave test computations by different schemes on the rectangular grid with h_y/h_x = ½. The density profiles obtained without artificial viscosity (left plot) and with the sAV and AV approaches (other plots).

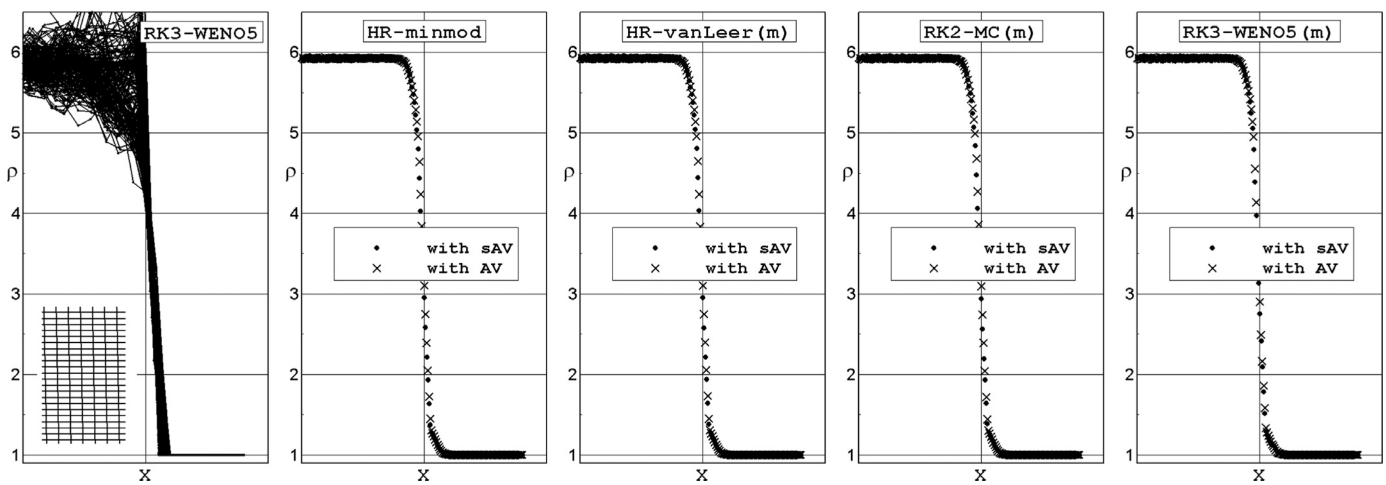

Fig. 5. Steady-shock-wave test computations by different schemes on the parallelogram grid with h_y/h_x = ½ and α = 0.02. The density profiles obtained without artificial viscosity (left plot) and with the sAV and AV approaches (other plots).

obtained using the original schemes and with the addition of the AV approach).

Thus, our study has revealed that the sAV approach suppresses the carbuncle instability and reduces the post-shock oscillations, much as the original AV approach does. Therefore, in the following discussions we will routinely use the common term “AV approach” for its both original and simplified forms.

4. Two approaches for curing the carbuncle instability

As already mentioned in Introduction, the common practice for curing the carbuncle instability consists in modifying the Riemann solver in use. Ideally, such modification must properly increase the numerical dissipation within the shock layer but it should not deteriorate the solution anywhere else (in particular, the grid-aligned contact discontinuity should not be smeared). The author has already given a summary on this subject in [7]; to this we can add recent publications [8,17–22], where some new modifications to the popular Riemann solvers were presented. This kind of curing the carbuncle phenomenon will be referred to as the Riemann Solver Modifying (or RSM for short) approach.

The AV approach was originally presented as an alternative to the RSM approach. The prime distinctive feature (and an essential advantage) of the AV approach is that it is external with respect to a specific Riemann solver. The added dissipative terms of the original AV approach in some ways resemble physical viscosity terms, and their form differs fundamentally from the form of inviscid fluxes. However, in the new (simplified) AV approach we refused to follow the right hand side of Navier–Stokes equations exactly, and the dissipative terms became much simpler. When looking at their modified form Eqs. (6) and ((7)) one may question whether these terms can be treated as a correction to the fluxes defined by the Riemann solver, and the sAV approach is a kind of the RSM approach. Let us give some considerations that help answer this question.

1. Even after its substantial simplification the AV approach keeps its basic features; thus, it does not affect the continuity equation. At the same time, the techniques based on the RSM approach make generally no exception for this equation and influence the intercell mass flux.
2. Let us consider the cell face that falls inside the shock layer, where the sAV approach implies adding the dissipative terms. In the RSM approach, the intercell flux is expressed as $\mathbf{F} = \mathbf{F}^{RS}(\mathbf{Q}_L, \mathbf{Q}_R, \mathbf{S})$, where $\mathbf{S}$ is the area vector of the cell face, $\mathbf{Q}_L$ and $\mathbf{Q}_R$ are the data states left and right of the interface, respectively. Suppose the flow across the cell face is supersonic and satisfying the condition $\min(S_L, S_R) > 0$, where S_L, S_R are the fastest signal velocities arising from the solution of the Rie-

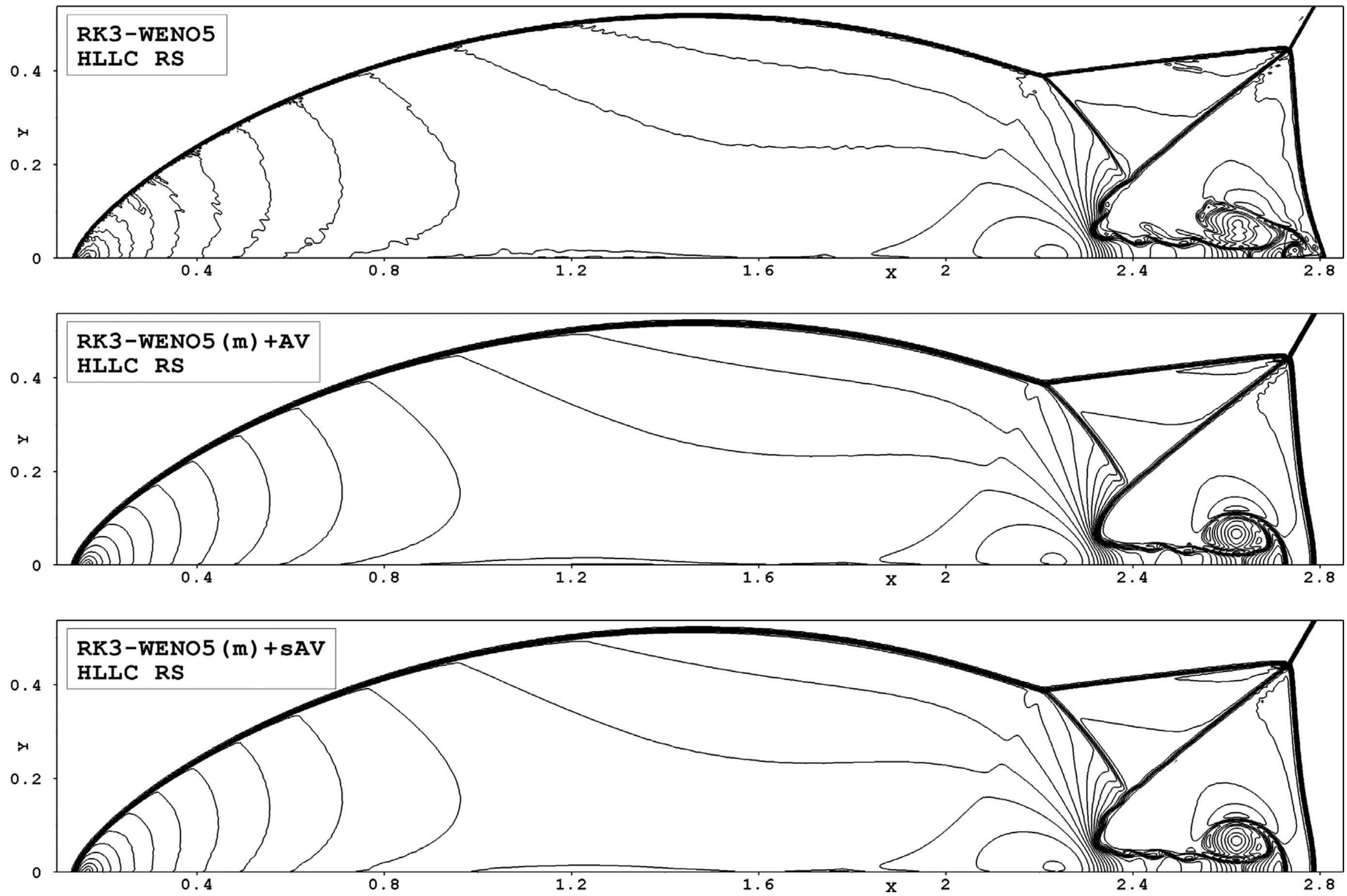

Fig. 6. Double Mach reflection problem. Forty one density contours equally spaced from 2.1 to 22. Computations by the RK3-WENO5 scheme on the grid with $h = 1/330$. Top: the original scheme, middle: with adding the original AV approach, bottom: with adding the simplified AV approach.

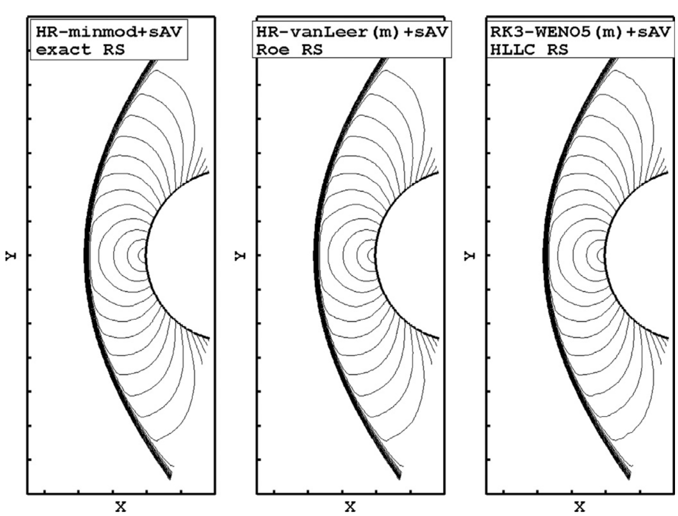

Fig. 7. Supersonic flow around a cylinder at $M_\infty = 3$. Twenty five equidistant Mach number contours from 0.1 to 2.5.

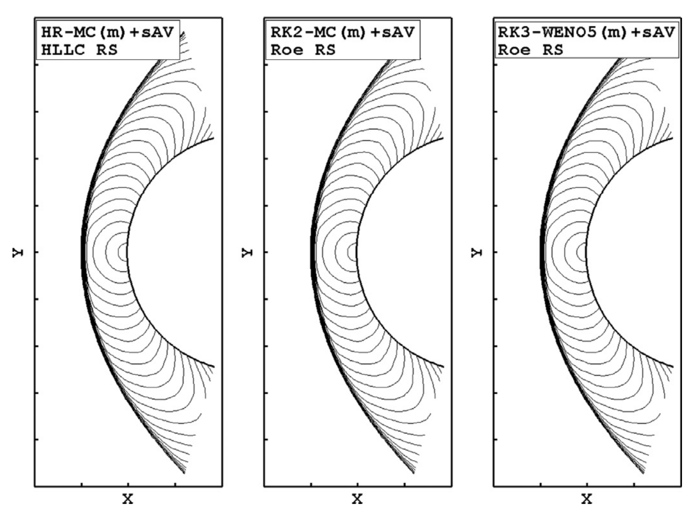

Fig. 8. Hypersonic flow around a cylinder at $M_\infty = 20$. Twenty five equidistant Mach number contours from 0.1 to 2.5.

mann problem. Any Riemann solver then computes the intercell flux by using the left data state only, that is $\mathbf{F} = \mathbf{F}(\mathbf{Q}_L, \mathbf{S})$. In this case, however, introducing any additional terms to $\mathbf{F}$ seems to contradict the basic principles of the RSM approach.

3. Unlike first-order schemes, in a high-order scheme, the data states $\mathbf{Q}_L$ and $\mathbf{Q}_R$ that are used in the RSM approach depend on the procedure of spatial reconstruction. However, the sAV approach (like the original AV approach) does not use these data states; it computes the dissipative terms uniformly, regardless of the particular reconstruction that appears in the high-order scheme.
4. When using any of the Runge–Kutta methods for marching the solution in time, one solves the Riemann problem at a particular cell face repeatedly (using updated data states $\mathbf{Q}_L$ and $\mathbf{Q}_R$ at each stage). In the case of using the HR scheme, the Riemann problem does not appear at the predictor stage, but it is solved at the corrector stage. In contrast, the sAV approach

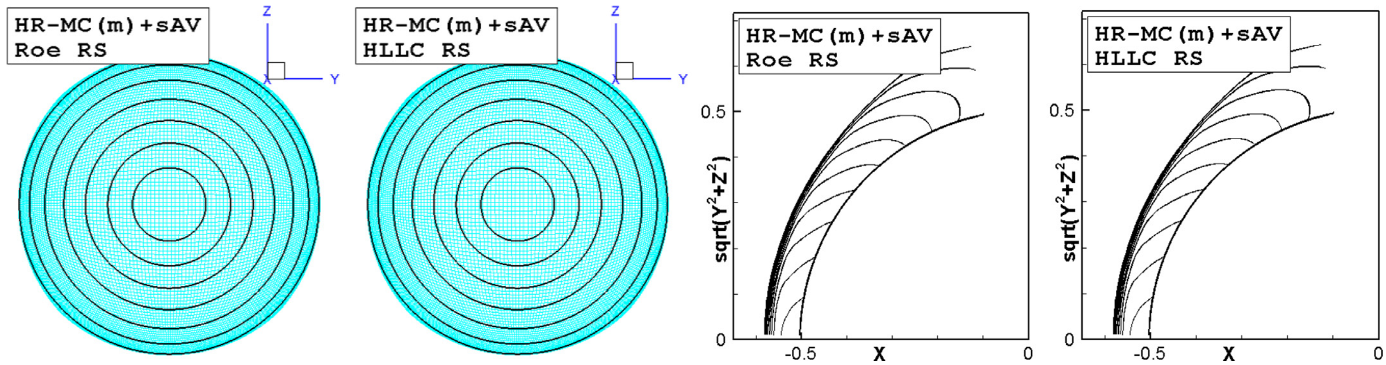

Fig. 9. Results for the Mach 10 flow around a sphere by the HR-MC scheme with the Roe and HLLC solvers. Ten Mach number contours equally spaced from 0.25 to 2.5 on the spherical surface $r = 0.55$ (left; view in yz plane) and on all grid surfaces j = constant (right; view in cylindrical coordinates).

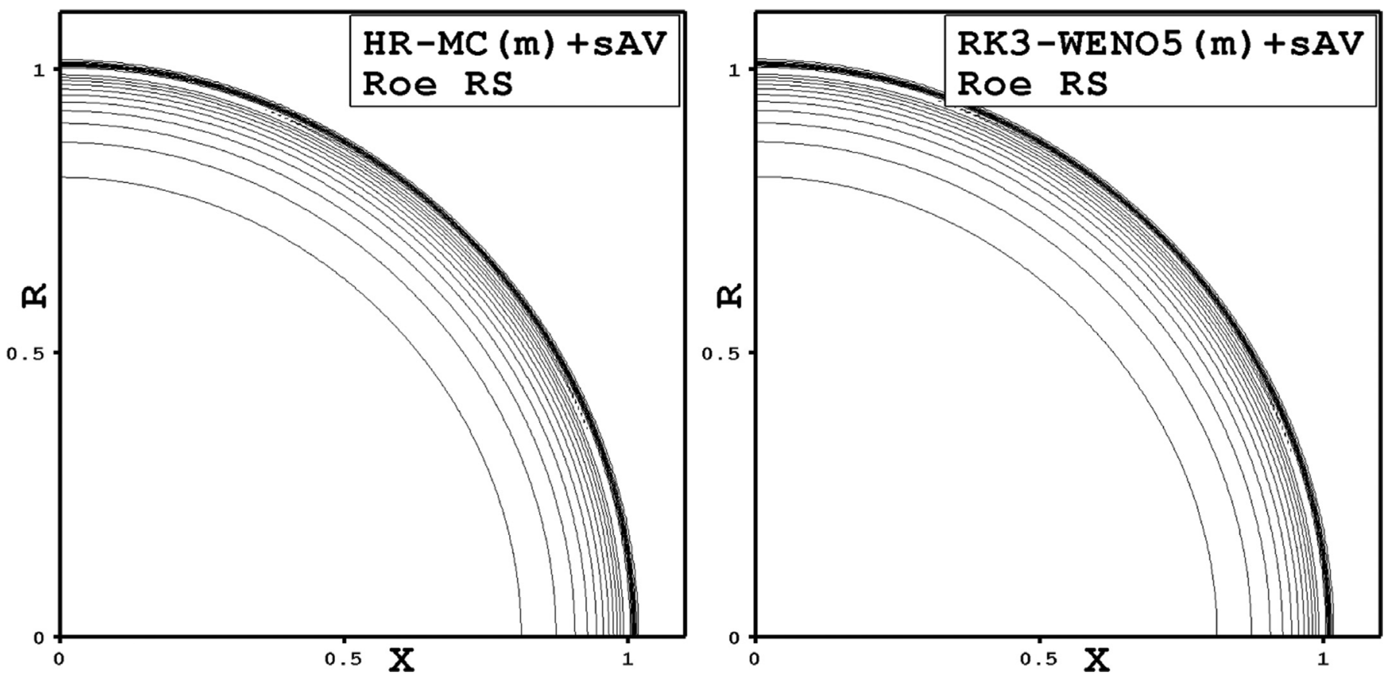

Fig. 10. Sedov blast at $t = 1$. Fifteen density contours equally spaced from 0.4 to 6.

(as well as the AV approach) implies computing the dissipative terms at the time level n and keeping them frozen at all stages of marching the solution in time.

These considerations lead to a definite answer to the above question that the dissipative terms of the sAV approach cannot be treated in the context of the RSM approach.

Thus, we have shown that the sAV approach differs structurally from the RSM approach. But these two approaches are supposed to be similar functionally: if they suppress the shock instability properly, they should demonstrate close results in a rich variety of test cases. It is of interest then to discover fine functional distinctions between the approaches under discussion. In what follows we disclose two of such distinctions that have been currently noticed. In so doing we confront the AV approach with a kind of the RSM approach that combines the Riemann solvers depending on the local flow conditions. In the latter technique, the HLLC solver (complete three-wave solver) is used as a basic one, whereas the HLL solver (two-wave solver) [23] or the Rusanov solver (one-wave solver) [24] is applied at the cell faces that fall inside the shock layer; we denote these combined solvers by HLLC-HLL and HLLC-Rusanov, respectively. In this kind of the RSM approach we compute the values of μ_{AV} as in the AV approach and use the condition $\mu_{AV} \neq 0$ to reveal that the cell face finds itself within the shock layer.

Distinction #1. Fig. 12 shows the computed results of the Quirk-type test – advancing shock wave on rectangular grid with $h_y/h_x = ¼$; the results have been obtained by the HR-MC and RK3-WENO5 schemes within the framework of the RSM approach. One can see that in this test case the RSM approach displays the shock-wave instability, as opposed to the sAV approach; compare Fig. 12 with Fig. 2. It goes along with our previous findings in [10]: applying an incomplete Riemann solver (thought to be free of the carbuncle instability) to the high-order Godunov-type schemes may not ensure that the shock instability is suppressed in simulations of the Quirk-type problem on a laterally compressed grid. In [10] we also gave an explanation for this failure: if a high-order scheme is used on a grid with $h_y/h_x \ll 1$, the lateral dissipation decreases substantially (recall that the procedure of spatial reconstruction reduces the difference between $\mathbf{Q}_L$ and $\mathbf{Q}_R$) promoting the instability even if an incomplete Riemann solver is applied.

Another example, in which similar situation occurs, is the kinked Mach stem problem from Quirk's catalogue [2]. In fact it is the double Mach reflection problem formulated in the following way. The computational domain in the xy plane is a rectangle [0, 1.5] × [0, 1] cut off by a straight line $y = (x – 0.075)\cdot\tan(30°)$; it is

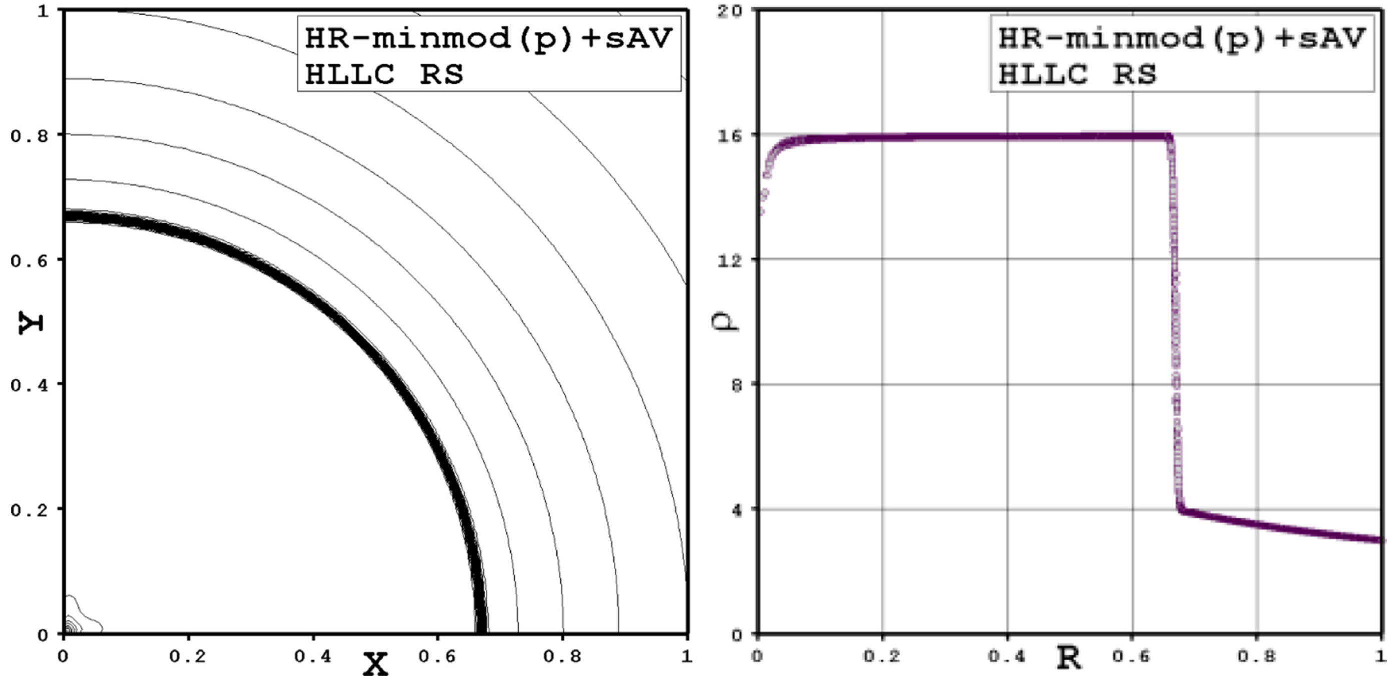

Fig. 11. Results for the Noh problem at t = 2. The density contour plots (from 2.5 to 17 with step 0.25) and the scatter plots of density versus radius.

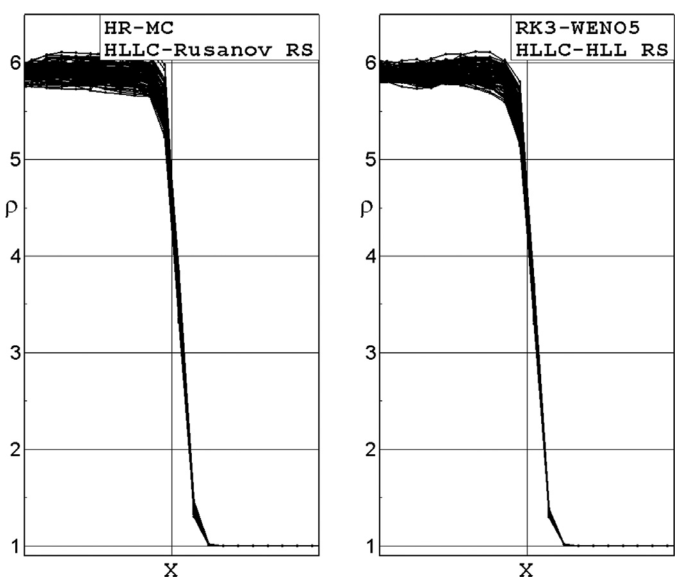

Fig. 12. Advancing-shock-wave test computations by the HR-MC and RK3-WENO5 schemes on the rectangular grid with h_y/h_x= ¼. The density profiles obtained within the framework of the RSM approach (HLLC-HLL and HLLC-Rusanov solvers).

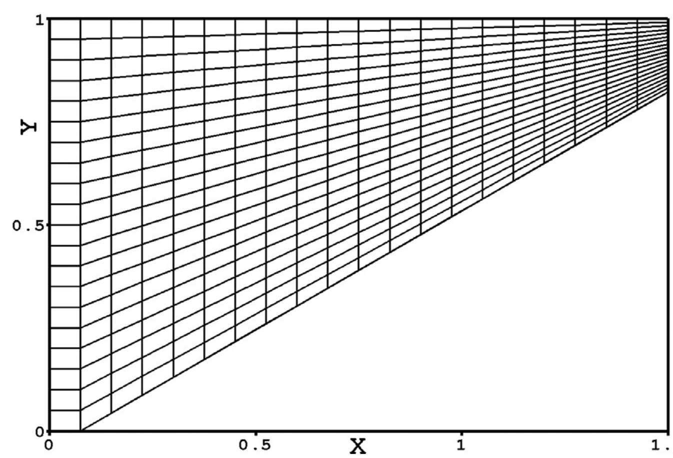

Fig. 13. Grid scheme in the kinked Mach stem problem.

covered with a regular grid composed of $J \times I = 800 \times 800$ cells, as shown schematically in Fig. 13. The initial conditions of a polytropic gas with $\gamma = 1.4$ in the domain are $(u_x,\ u_y,\ \rho,\ P) = (0,\ 0,\ 1,\ 1/\gamma)$. The bottom, upper and right boundaries are solid walls. The left boundary condition is the inflow corresponding to $M_S = 5.5$. The output time is $t = 1.2/M_S$.

Fig. 14 shows the results obtained by the HR-MC (top panel) and RK3-WENO5 (bottom panel) schemes within the framework of both the sAV (left plots) and the RSM (right plots) approaches. Although the Mach stem is not kinked in all the computations, one can clearly see that in this specific test case the RSM approach does not fully suppress the shock-wave instability, whereas the sAV approach does.

Distinction #2. Fig. 15 shows an example of using the RSM approach in the case of computing the double Mach reflection problem by the RK3-WENO5 scheme (here we turn back to the original formulation of the problem with the improvement discussed in [10]). Similar results, but for the sAV approach, were shown earlier in Fig. 6. A comparison between these figures reveals two distinctions: (1) the RSM approach demonstrates conspicuous oscillations behind the reflected shock wave; (2) the sAV approach suppresses such oscillations, but increases slightly the shock broadening. These findings can be commented in the following way.

In the context of the Euler equations, shock waves are treated as discontinuities, across which the Rankine–Hugoniot conditions are valid. Shock-capturing methods (schemes), however, smear the shock front over the computing cells, forming the shock layer. Although the Rankine–Hugoniot conditions are not explicitly applied in such computations, one can expect that they are approximated integrally (over the shock layer) if the shock-capturing method approximates the conservation laws. The evident intention to minimize the shock front smearing generally meets a restriction on the scale of spurious post-shock oscillations that occur when the Rankine–Hugoniot conditions are not sufficiently well approximated. Consequently, when adjusting the shock-capturing method (or one of its key components) for multidimensional problem sim-

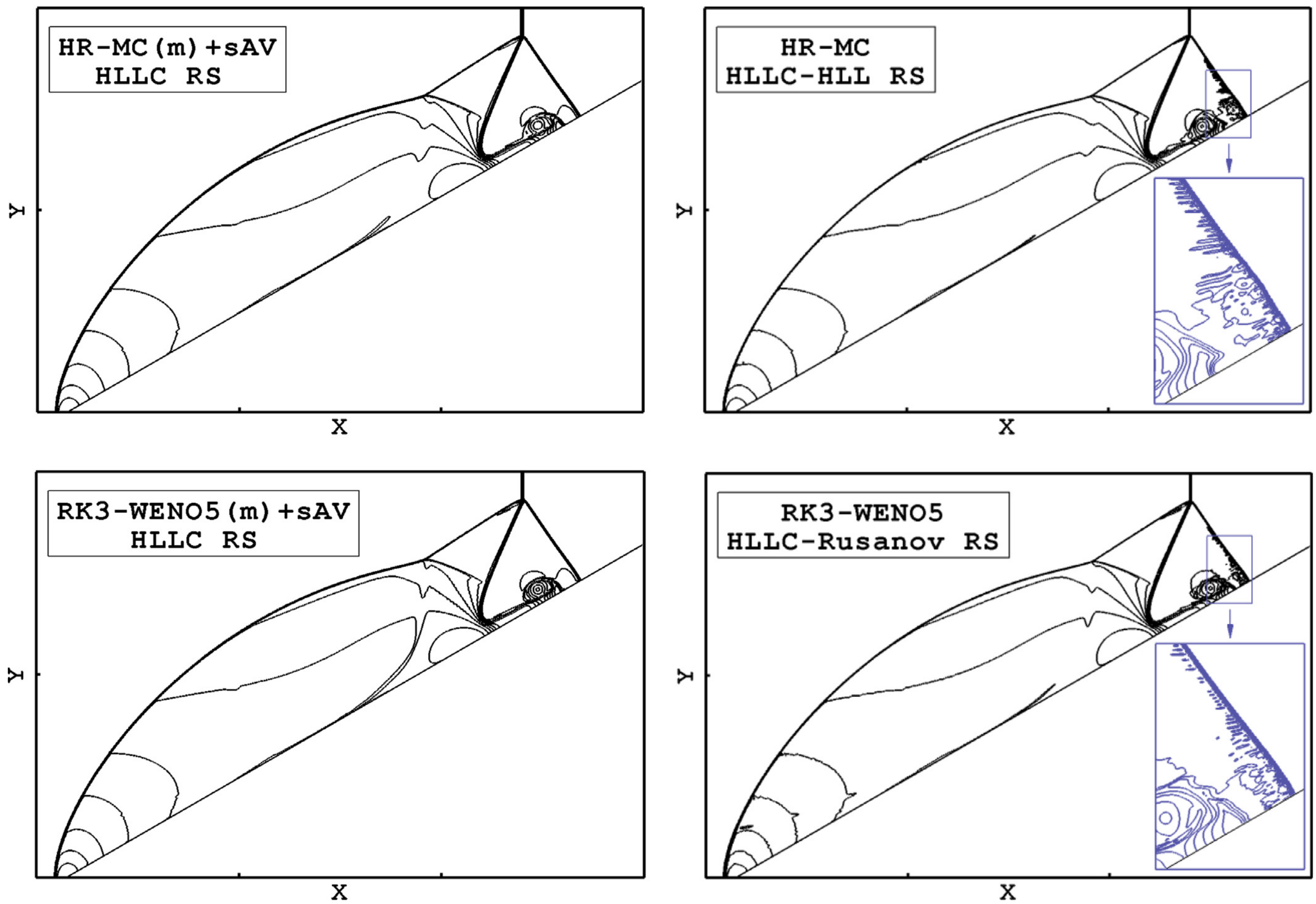

Fig. 14. Kinked Mach stem problem from Quirk's catalogue (variant of the double Mach reflection problem). Twenty one density contours equally spaced from 2 to 12. Left: HR-MC and RK3-WENO5 schemes with the sAV approach; right: respective schemes with the RSM approach.

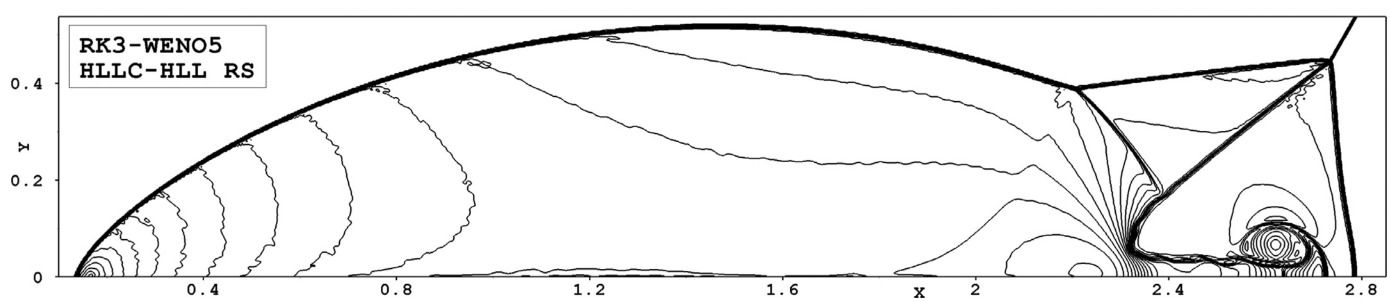

Fig. 15. Double Mach reflection problem. Forty one density contours equally spaced from 2.1 to 22. Computations by the RK3-WENO5 scheme on the grid with $h = 1/330$.

ulations, it is highly important to optimize the dissipative properties of the resulting scheme. In the test case under discussion, the artificial viscosity approach seems to be more balanced than the RSM approach.

5. The AV approach and Riemann solvers

At this point, let us recall that the added dissipative terms in the AV approach somewhat resemble the *viscous fluxes* across the cell faces that fall inside the shock layer. In this way the AV approach gains high efficiency and versatility: it is external with respect to the scheme that approximates the Euler equations and, in particular, to the Riemann solver that computes *inviscid fluxes*. Thus, preventing the shock instability, the AV approach allows us to revert to the plain opinion accepted in the past (before encountering the carbuncle phenomenon): physically, the exact Riemann solver represents the most consistent way to treat inviscid fluxes between cells. Now, having done away with the long-drawn-out problem, we may ask ourselves again the old question: which of the approximate Riemann solvers gives the best fit to the exact Riemann solver in the case of computing problems of one kind or another by Godunov-type schemes? In what follows we demonstrate evaluations of this kind for several approximate Riemann solvers as applied to a specially selected test problem: *shock wave moving in a stepped tunnel*. The problem formulation is as follows.

The domain in the xy plane is a rectangle $[0, 1] \times [0, 1]$ with two cut-offs: $[0, 0.1] \times [0, 0.5]$ and $[0.6, 1] \times [0.5, 1]$; it is covered with a square grid with spacing $h = 1/1000$ (normal grid) or $1/2000$ (fine grid). The initial conditions of the gas ($\gamma = 1.4$) in the domain are $(u_x, u_y, \rho, p) = (0, 0, 1, 1/\gamma)$. The upper part of the left boundary ($x = 0$, $0.5 < y < 1$) is the inflow corresponding to

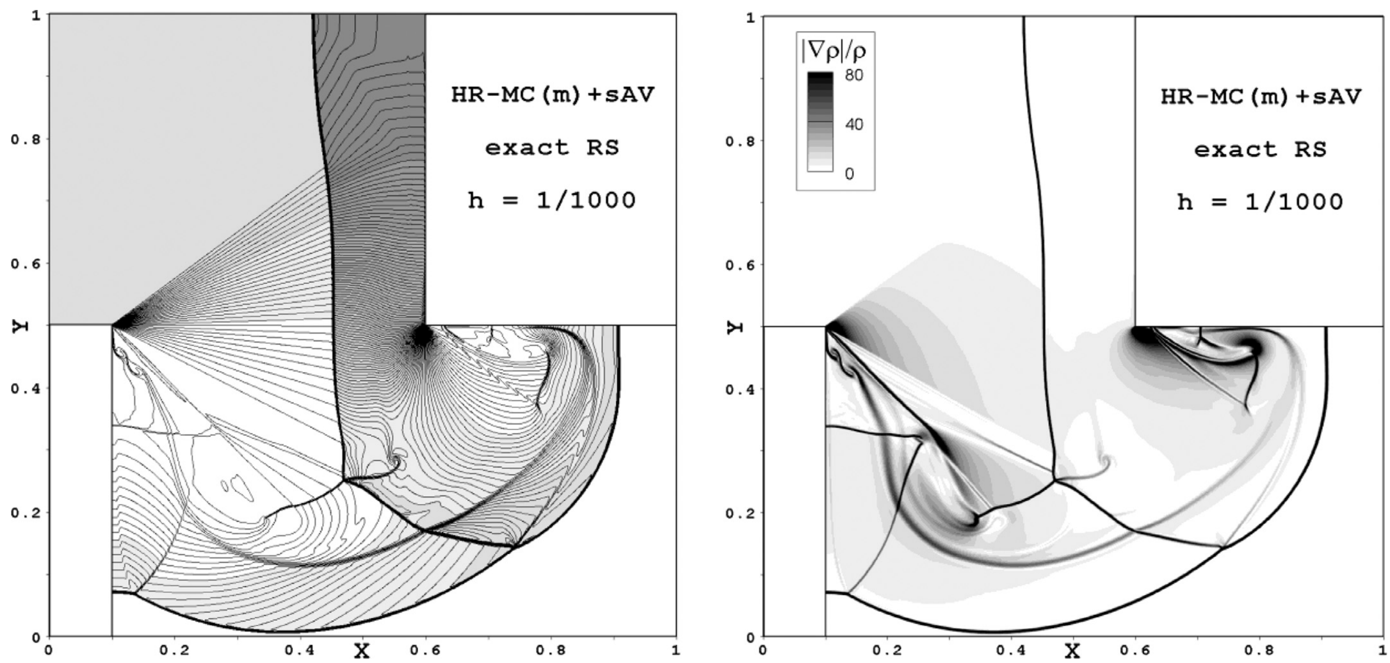

Fig. 16. Shock wave moving in a stepped tunnel. Left: 125 density contours equally spaced from 0.1 to 16; right: the numerical schlieren image.

$M_S = 5$; the rest of the boundaries are solid walls. The output time is $t = 0.22$.

When computing this test problem, we make a one-time correction: at time $t = 0.02$ (the shock wave reaches the first tunnel step at $x = 0.1$), the exact values of flow variables behind the Mach 5 shock are placed at all the grid cells that are located within the region $x < 0.1 - 5h$. The purpose of this is to suppress small spurious perturbations that arise during the process of forming the incoming shock just after $t = 0$ (within the first time steps); see section 15.8.4 of the book [11] about the "start-up errors".

Fig. 16 shows the numerical results obtained by the HR-MC(m)+sAV scheme with the exact Riemann solver and the grid spacing $h = 1/1000$. As one can see, this test problem considers a complex flow structure that includes regions of strong compression and rarefaction, contact discontinuities and shock waves interacting with each other. As the flow parameters in this test are very sensitive to both the numerical scheme and the grid, it is well-matched to the objective of our study – to evaluate the influence of changing the exact Riemann solver (that gives the reference solution) for its approximate alternatives. For the approximate solvers we select several versions of the Roe and HLLC solvers, which will be specified later.

We compute the local deviation of one solution from another by

$$\varepsilon = \frac{2\left|\rho^{(1)} - \rho^{(2)}\right|}{\rho^{(1)} + \rho^{(2)}} \cdot 100\%,$$

where $\rho^{(1)}$ and $\rho^{(2)}$ are the local densities in two solutions of the problem.

Fig. 17 shows the maps of deviations from the reference solution illustrated in Fig. 16. The *left top plot* compares the solutions obtained on the two grids (fine grid vs. normal grid) by the same scheme and the exact Riemann solver. This map illustrates how sensitive the solution is to the grid refinement. One can see that in the upper region ($y > 0.5$) the numerical errors nowhere exceed 0.3% except the smeared shock front (for obvious reasons) and a small part of the upper boundary of the rarefaction wave located near the first tunnel step (the spatial derivatives of the flow variables are discontinuous there). In the lower region ($y < 0.5$), the errors are markedly higher. Here, in addition to the shock fronts one can see several contact discontinuities, the sharpest of which reveal instability. This instability heavily affects the flow in two rarefaction zones located below the first and second tunnel steps; in the left top plot they are marked by *zone A* and *zone B*.

Consider now the *right top plot* that compares the solutions obtained with the Roe approximate Riemann solver and the exact solver. Here, a noticeable decrease in accuracy in the lower part of the rarefaction wave (arrow 1) is evident. In zone A, the deviation from the reference solution as a whole remains within the grid sensitivity levels (shown in the left top plot), but in zone B, the errors exceed these levels abundantly. The *left middle plot* shows the deviation map for the Roe solver supplemented with Harten's entropy fix [25]. It can be seen that this plot looks nearly identical to the right top plot. This fact proves the opinion that the entropy problem of the first-order scheme with Roe's solver (Fig. 18 illustrates this problem) fades away after switching to its high-order extensions.

The *right middle plot* of Fig. 17 shows the effect of changing the exact Riemann solver for the HLLC solver, more precisely, for its basic variant described in section 10.6 of the book by Toro [26]. This plot reveals the following. Compared to the Roe solver, using the HLLC solver decreases the errors almost everywhere: in the lower part of the rarefaction wave and in zones A and B. The only exception is a small area below the second tunnel step (arrow 2), where the deviation may amount to 0.5 %.

The deviation map shown in the *left bottom plot* of Fig. 17 relates to a simplified version of the HLLC-type solvers that was formulated by Safronov [30]. We denote this solver by sHLLC (*s* stands for simplified); Appendix A gives the reader a brief summary of it. Looking at this map, one may notice a stronger manifestation of the reflected shock wave (arrow 3) than what was observed for the solvers discussed above. This effect is due to a moderate difference from the reference solution in one or two central cells of the shock layer (where the gradients are at maxima), whereas the flow parameters at other points of the shock layer, and accordingly its width, correspond perfectly well to the reference solution. The latter suggests that the effect under discussion is of no practical importance. In zone B, the errors are nearly the same as those found for the basic HLLC solver, but in zone A we observe that the deviation generally decreases. Since the sHLLC solver represents a

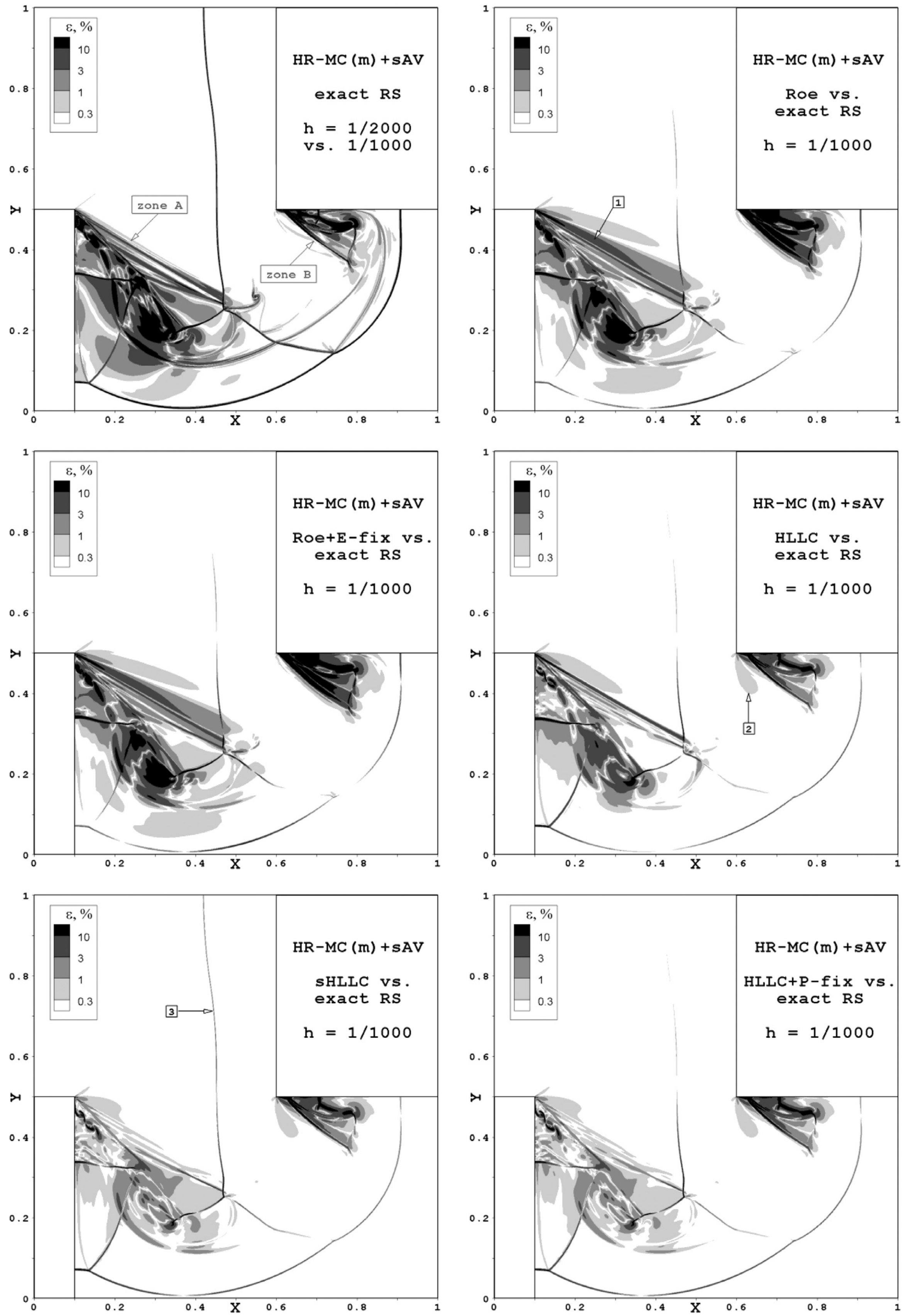

Fig. 17. Shock wave moving in a stepped tunnel. The local deviation maps for a variety of solutions.

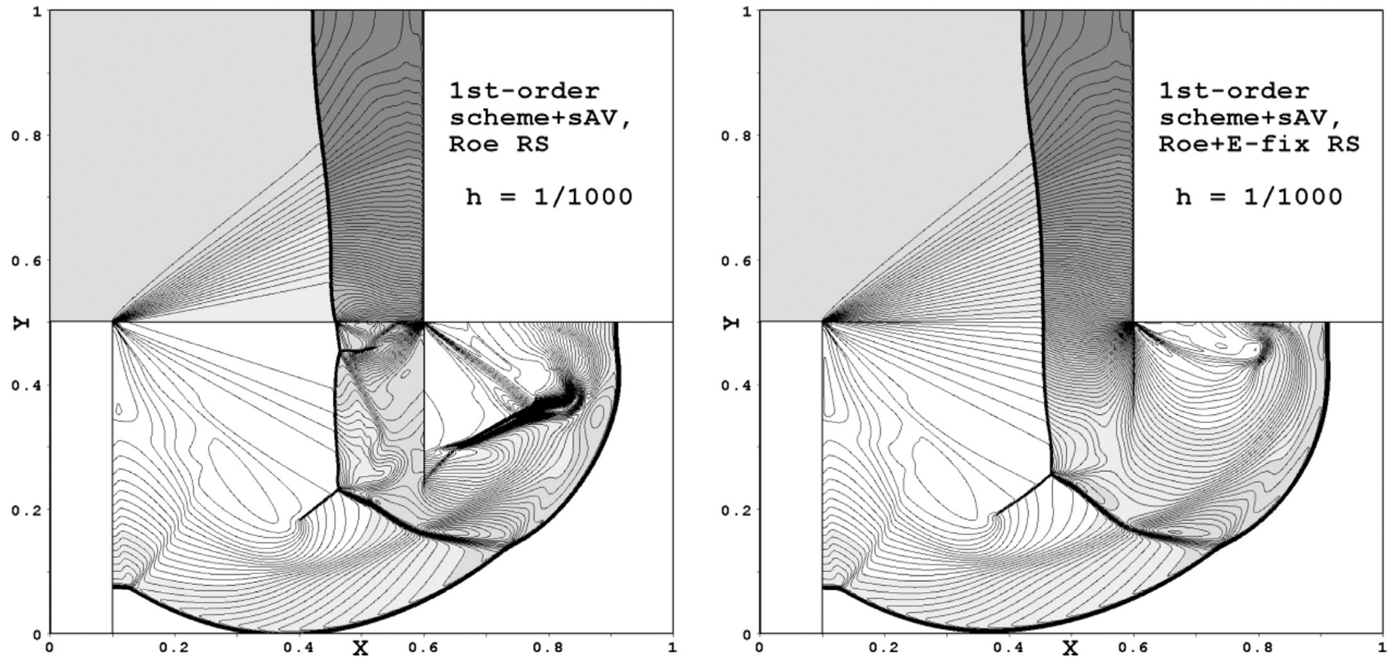

Fig. 18. Shock wave moving in a stepped tunnel. 125 density contours equally spaced from 0.1 to 16. Computations by the first-order Roe scheme without (left plot) and with (right plot) the Harten entropy fix.

simplified version of the HLLC-type solvers, this result needs clarification.

There are two distinctions between the basic and simplified versions of the HLLC solver. First, they differ in the wave speed estimates for S_L and S_R: the basic version uses more refined algorithm that provides better agreement with the reference solution in the core of shock layer (the meaning of this effect has just been discussed). Second, the basic HLLC solver calculates the resulting flux by the use of conserved variable vectors, whereas the sHLLC solver is formulated in terms of the primitive variables. These two algorithms are functionally equivalent except when the restriction in (B.8) is activated, preventing the pressure from being negative. The *right bottom plot* of Fig. 17 shows the deviation map for the basic HLLC solver supplemented with the same restriction, called a *pressure fix*. One can see that the pressure fix does not affect the flow in zone B, and yet it reduces the errors in zone A to the level of the sHLLC solver.

From these findings we conclude that the sHLLC solver, being coupled with the AV approach, can provide benefits in complex practical applications.

6. Conclusions

In this work we presented a simplified form of the artificial viscosity approach (the sAV approach) for curing the carbuncle phenomenon and suppressing post-shock oscillations. If the original AV approach introduces additional dissipation in the form of the right-hand side of Navier–Stokes equations, the new sAV approach adds the dissipative terms in a more concise form. The extensive testing on carbuncle-related problems has demonstrated that the new approach is comparable to the original one in its efficiency. Thus we can recommend the sAV approach for use in codes developed solely for solving the Euler equations.

Moreover, we discussed the distinctive features of the AV approach (in both its original and simplified forms) as compared to the other way of curing the carbuncle phenomenon by modifying the Riemann solver in use (the RSM approach). In particular we argued that the dissipative terms of the sAV approach cannot be treated in the context of the RSM approach. We also revealed and illustrated two functional distinctions between the AV and RSM approaches.

Finally, we computed the test problem of a shock wave moving in a stepped tunnel and evaluated the effect of changing the exact Riemann solver (that gives the reference solution) for one of its approximate alternatives: the Roe-type or HLLC-type solver. On the whole, the solutions obtained with the HLLC-type solvers were found to provide the best fit to the reference solution. We also concluded that the simplified HLLC solver, being coupled with the AV approach, can provide benefits in complex practical applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Alexander V. Rodionov: Conceptualization, Methodology, Software, Investigation, Writing, Visualization.

Acknowledgment

The author would like to thank Tatiana Zezyulina for the qualified assistance in English and the reviewers for their helpful comments.

Appendix A. Inviscid strong vortex-shock wave interaction

This problem has been proposed to participants of the 5th International Workshop on High-Order CFD Methods (held in Kissimmee, FL, USA, 6-7 January 2018) as an advanced test case for assessing the performance of high-order shock-capturing methods (order of accuracy > 2). In particular, it was discovered in the in-depth study (see the related presentations in the web site [27]) that all the computations made suffered from conspicuous numerical artifacts – post-shock oscillations appearing after the vortex bends the shock front (which is initially plane). In this appendix we demonstrate that the artificial viscosity approach can significantly suppress these artefacts.

Problem formulation. Hereinafter we consider a polytropic gas with $\gamma = 1.4$. The computational domain $[0, 2] \times [0, 1]$ in the xy plane is covered with a regular square mesh of size h. Initially, a stationary shock with shock-wave Mach number $M_S = 1.5$ is located at $x = 0.5$, and a strong isentropic vortex with $M_v = 0.9$ is centered at $(x_c, y_c) = (0.25, 0.5)$. Hence, the flow field in the region $x < 0.5$ is a superposition of two flow patterns: a *mean upstream flow* with $(u_x, u_y, \rho, p) = (u_S, 0, 1, 1)$, with $u_S = M_S\sqrt{\gamma}$, and an *isentropic vortex* (perturbation of the mean upstream flow). The vortex rotates counter-clockwise with the angular velocity defined as

$$u_\theta = \begin{cases} u_m \frac{r}{a} & \text{if } r \le a, \\ u_m \frac{a}{a^2 - b^2}\left(r - \frac{b^2}{r}\right) & \text{if } a < r < b, \\ 0 & \text{if } r \ge b, \end{cases}$$

where r is the distance from the vortex core (x_c, y_c), $u_m = M_v\sqrt{\gamma}$ is the maximum angular velocity (at $r = a$), and $(a, b) = (0.075, 0.175)$. The *downstream flow* conditions ($x > 0.5$) are defined from the values of the mean upstream flow and M_S. The left and right boundaries are considered as a supersonic inlet and a subsonic outlet, respectively. The bottom and upper boundaries are solid walls. The output time is $t = 0.7$. More detailed information on the initialization process can be found in [27].

Numerical artifacts. First we recall that the carbuncle instability is not appearing at $M_S = 1.5$. But the test case under study has to do with another problem from Quirk's catalogue [2], namely the problem of computing a slowly moving shock [28]. Actually, after the vortex reaches the shock front, the latter starts to move slowly through the grid in a two-dimensional manner. Spurious low-frequency perturbations in the downstream flow are much expected in such a case.

Fig. A.19 shows the numerical results obtained by the RK3-WENO5 scheme with the grid spacing $h = 1/600$ (top plots) and $h = 1/900$ (bottom plots); the HLLC Riemann solver was used within this study. The schlieren images (here $Sch = \ln(1 + |\nabla\rho|)/\ln 10$) obtained without adding the artificial viscosity approach to the basic scheme (left plots) reveal strong post-shock oscillations of long wavelengths (~15h). When the artificial viscosity is applied, these oscillations appear to be substantially damped

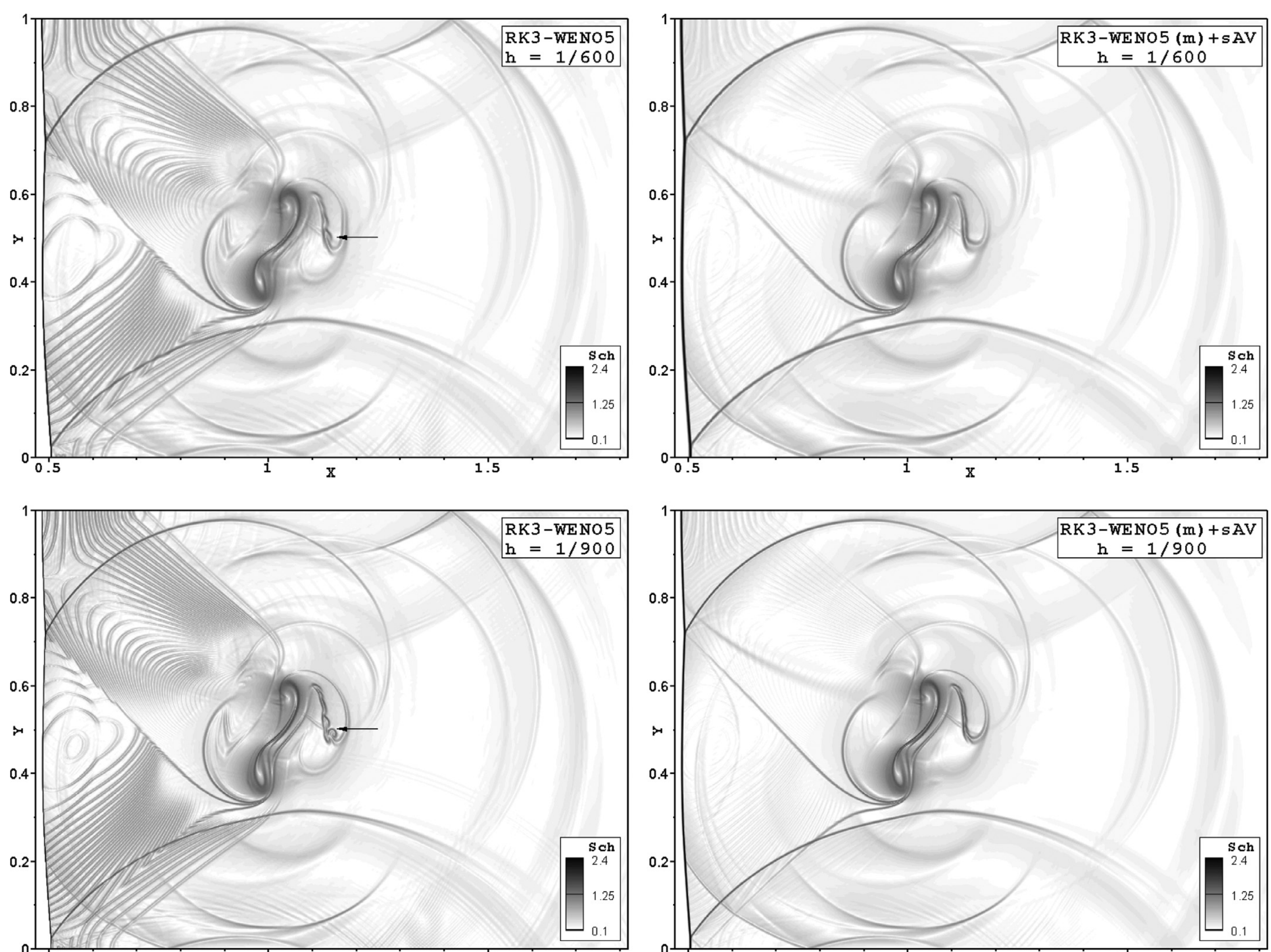

Fig. A.19. Strong vortex-shock wave interaction. The numerical schlieren images obtained by the RK3-WENO5 scheme in its original form (left) and with the sAV approach (right) on the grid with $h = 1/600$ (top) and $h = 1/900$ (bottom).

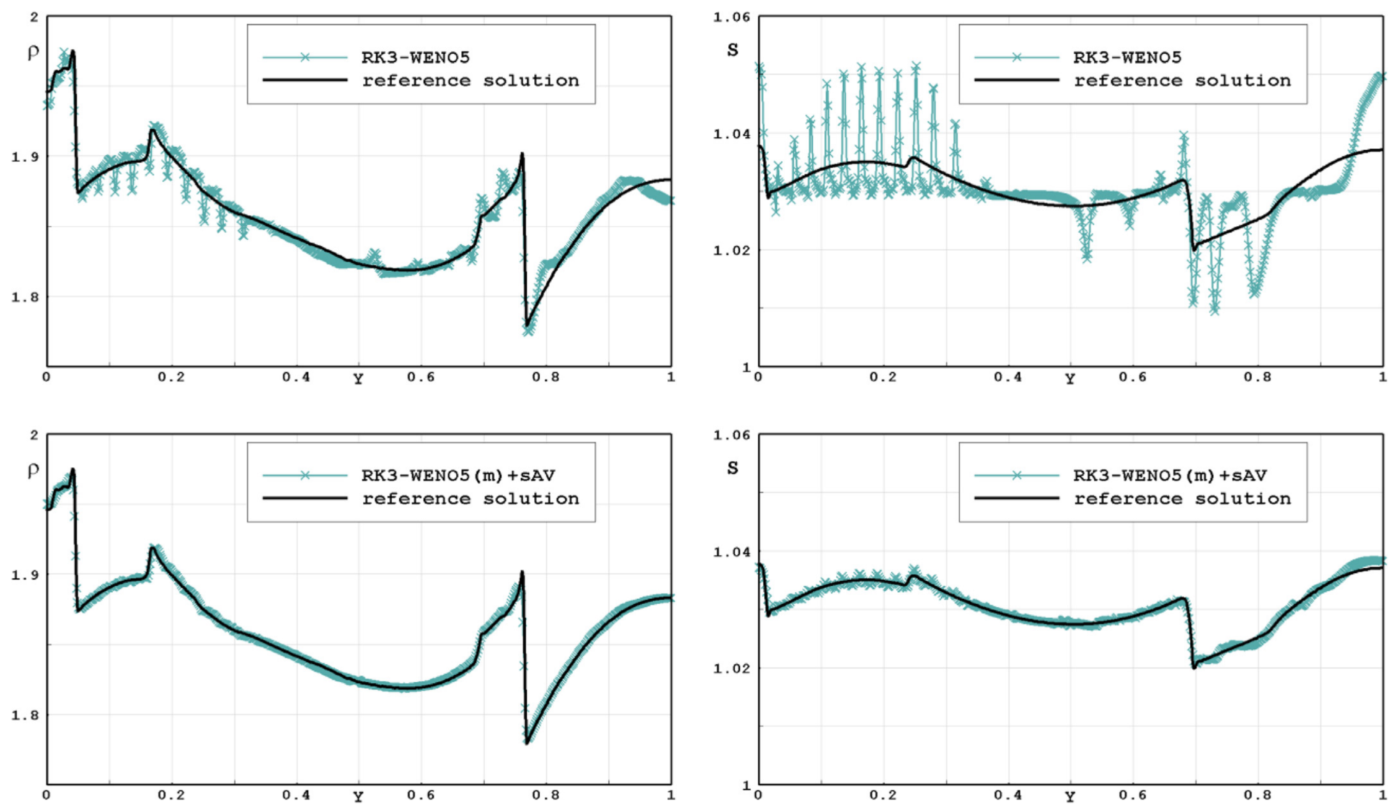

Fig. A.20. Strong vortex-shock wave interaction. The profiles of the density (left) and entropy (right) at $x = 0.52$ as compared with the reference solution. The data obtained by the RK3-WENO5 scheme in its original form (top) and with the sAV approach (bottom) on the grid with $h = 1/600$.

– they are barely visible in the right plots of Fig. A.19. Besides, the RK3-WENO5 scheme in its original form demonstrates some kind of instability ahead of the split vortex (see arrows in the left plots), whereas being completed with the sAV approach it provides stable solution.

Fig. A.20 shows the profiles of density (left plots) and entropy ($S = p/\rho^\gamma$; right plots) at $x = 0.52$ (just behind the leading shock) as compared with the reference solution (to be described later). Here we examine the results obtained with the grid spacing $h = 1/600$. The top plots demonstrate the results by the original RK3-WENO5 scheme; they are burdened with severe perturbations, of which the most pronounced effect occurs in the entropy. Adding the artificial viscosity (bottom plots) yields a large dividend in quality of the solution. Thus, the amplitude of artificial oscillations in entropy decreases by about an order of magnitude.

Reference solution. We have used the RK3-WENO5 scheme with the sAV approach and the grid spacing $h = 1/1200$ to obtain the reference solution. A crucially new practice here is that we reformulated the problem, subtracting the value of u_S from the x–component of velocity in both the upstream and downstream flow conditions. So, in the new formulation the mean upstream flow becomes motionless, and the leading shock wave is moving to the left; in so doing we avoid computing a slowly moving shock. Besides, to save computing resources we have started the computation with the domain $0 \le x \le 0.5 + 20h$ and then, as needed from time to time, we lengthened the domain on the left and shortened it on the right in segments of $20h$. When saving the computational results we recovered the velocity and the grid coordinate by: $u_x + u_S \to u_x$, $x + u_S\, t \to x$.

It is pertinent to discuss the reference solution in some detail. Fig. A.21 shows the numerical schlieren images as well as the entropy and vorticity maps (the vorticity $\omega = \partial u_y/\partial x - \partial u_x/\partial y$) at times $t = 0.15$, 0.25, 0.35, 0.5 and 0.7. By inspecting these plots one can reveal the origins of distinct flow discontinuities and follow their evolution. Thus, at $t \ge 0.25$ the reflected shocks (arrows 1 in the top plots) and the contact discontinuities (arrows 2 in the bottom plots) that emerge from the triple points (circled in the plots) are clearly visible. One can also note the acoustic waves that emanate from the strongly disturbed vortex; some of them become the weak shocks (for instance, see arrows 3 in the top plots). The entropy maps (middle plots) are highly informative as well. Indeed, both the upstream and downstream flows are initially isentropic, but after the vortex distorts the shock the downstream flow ceases to be isentropic. Moving out of shock waves, each fluid particle preserves entropy. Therefore by tracking the movement of entropy contours we can monitor the flow of gas. The middle plots show the entropy map in a gray scale and, additionally, the contours $s = 1.018$. Thus, the selected contours clearly show how the initially single mass of fluid is torn apart. As this takes place, extremely narrow bridges remain between the individual parts of the fluid; these bridges, however, erode due to the numerical dissipation. The effect of this kind can also be revealed by tracking the regions $s > 1.12$ (black flooding).

From the aforesaid it follows that computing the test case under discussion is a challenging task.

Remark. When computing this test problem, the following procedure has been used to remove the start-up errors that arise in the course of smearing the initial shock front (and forming the shock layer). At time $t = 0.075/u_S$ (the vortex reaches the shock wave), the exact downstream values were reset at all the grid cells behind the shock layer.

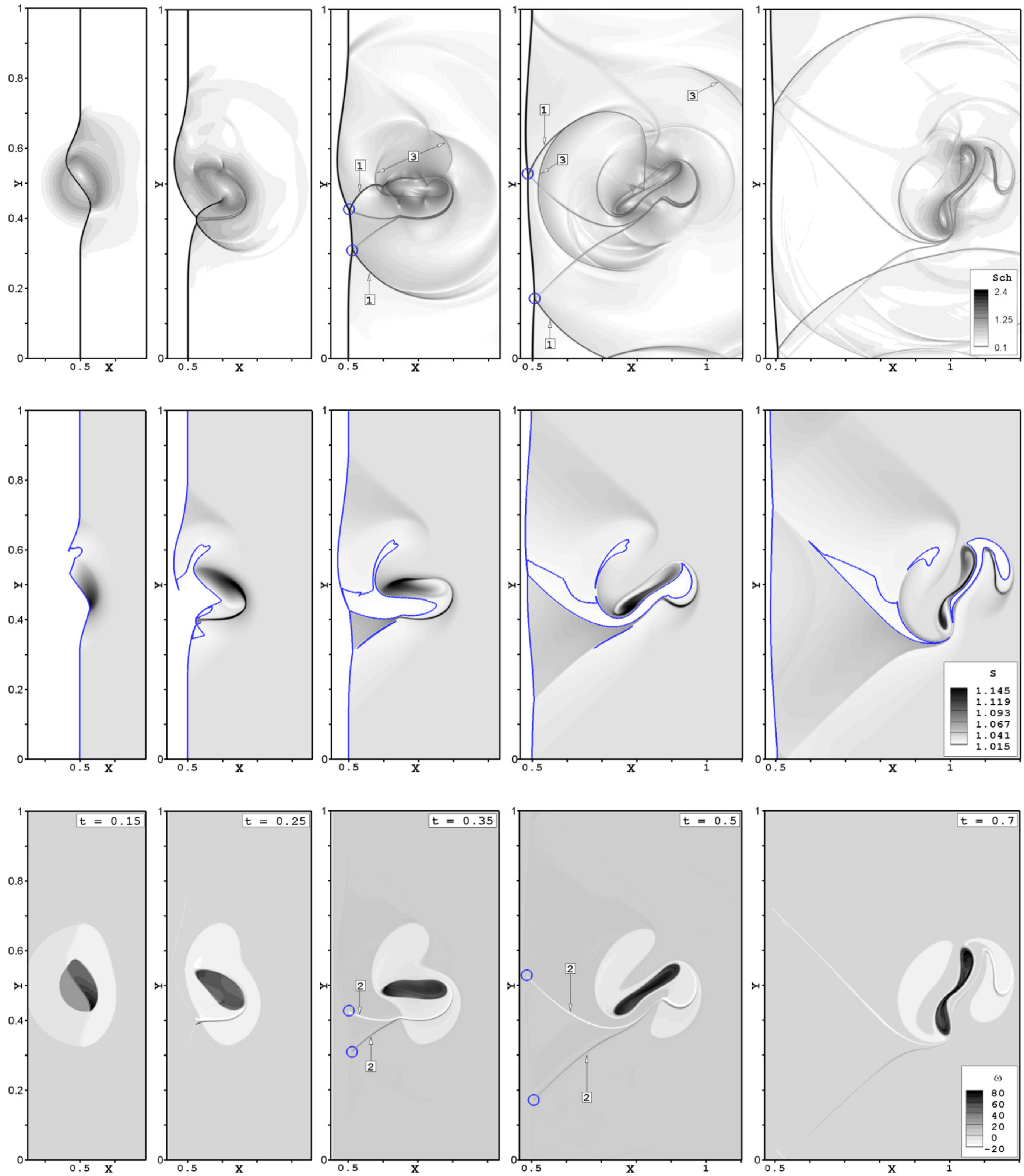

Fig. A.21. Strong vortex-shock wave interaction. The reference solution at times t = 0.15, 0.25, 0.35, 0.5 and 0.7 (from the left to the right). The numerical schlieren images (top), the entropy mars (middle) and the vorticity maps (bottom).

Appendix B. Simplified HLLC solver

This appendix provides a brief summary of the HLLC-type approximate Riemann solver as formulated by Safronov [30]. In what follows we will deal with a gas, for which its specific enthalpy h (or internal energy $e = h - p / \rho$) and the sound speed a are known functions of p and ρ.

Let us consider the Initial Boundary Value Problem for the two-dimensional Euler equations. The initial discontinuity at the interface $x = 0$ can be formulated as

$$\mathbf{Q}(x, t=0) = \begin{cases} \mathbf{Q}_L & \text{if } x < 0, \\ \mathbf{Q}_R & \text{if } x > 0, \end{cases}$$

where $\mathbf{Q} = [u, v, \rho, p]^T$ is the vector of the primitive variables. Our objective is to evaluate the flux vector $\mathbf{F}_x = [\rho u, \rho u^2 + p, \rho uv, \rho u h_0]^T$ at $x = 0$, $t > 0$.

At first, using the components of the known vectors $\mathbf{Q}_L$ and $\mathbf{Q}_R$ we compute the specific total enthalpies $(h_0)_L$ and $(h_0)_R$ (required for evaluating $\mathbf{F}_x$) as well as the sound speeds a_L and a_R. Then, we use the estimate suggested by Davis [29] to compute the fastest signal velocities, namely

$$S_L = \min\{u_L - a_L, u_R - a_R\}, S_R = \max\{u_L + a_L, u_R + a_R\}.$$

By assuming that the surfaces $x/t = S_L$ and $x/t = S_R$ are discontinuities (compression or rarefaction shocks), across which the general laws of conservation of mass, momentum and total energy are valid, we may derive all the values in the *Star Region* (the term taken from the book [26]). The solution algorithm can be formulated in terms of the primitive variables as follows:

(1) Compute the mass fluxes (positive values) across the discontinuities as

$$m_L = \rho_L(u_L - S_L), m_R = \rho_R(S_R - u_R).$$

(2) Compute the pressure and the normal velocity component as

$$p_{*L} = p_{*R} = p_* = \max\left\{0, \frac{p_L m_R + p_R m_L + m_L m_R (u_L - u_R)}{m_L + m_R}\right\}, \quad \text{(B.8)}$$

$$u_{*L} = u_{*R} = u_* = \frac{u_L m_L + u_R m_R + p_L - p_R}{m_L + m_R}.$$

(3) Compute the other values as

$$v_{*L} = v_L, \rho_{*L} = m_L/(u_* - S_L), (h_0)_{*L} = (h_0)_L + S_L(u_* - u_L),$$
$$v_{*R} = v_R, \rho_{*R} = m_R/(S_R - u_*), (h_0)_{*R} = (h_0)_R + S_R(u_* - u_R).$$

(4) Evaluate the flux vector $\mathbf{F}_x$ based on the values of the primitive variables (supplemented by h_0) that fall on the surface $x = 0$ (in accordance with the signs of three speeds: S_L, u_* and S_R).

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